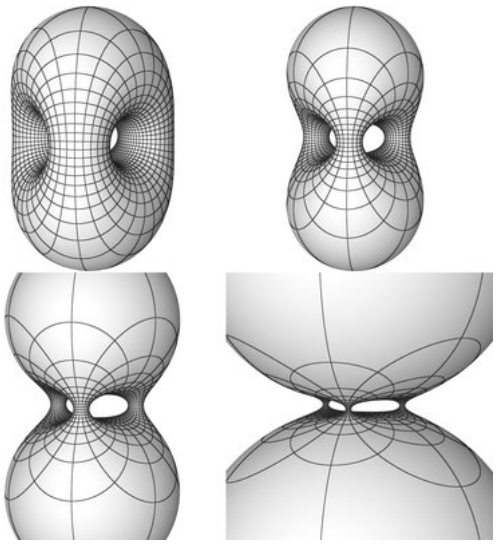
Riemann Surfaces : a Glance



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2025 Spring

Introduction

This is a note about the theory of Riemann surfaces, which can not be very complete due to the limited knowledge of the author about algebraic geometry and complex geometry (and number theory). I write this note to make the theory clearer to me essentially, and I would like to see someone else getting interested in the theory of Riemann surfaces after reading this **incomplete** note. If there are such readers in Qiuzhen Colledge, I would like to hold a seminar discussing these topics.

Riemann started the study of Riemann surfaces (this is why these surfaces are named after Riemann) by defining and studying "the surface of a complex function" in order to find the most natural way to continuate a function. Since then many excellent mathematicians have worked in this area. For example, the modern definition of a manifold is first given in Weyl's book *The Concept of a Riemann Surface*. Till now, the study of complex structures on surfaces, which is called **Teichmuller theory**, is still an active direction of modern mathematics.

We will first start by giving basic definitions of Riemann surfaces and then give a quick introduction to sheaves.Anyone that don't want to read materials about sheaves can just omit section 1.2, 1.3, 2.2 and 3.4 without causing any problem while reading the rest.

In chapter 2 we study the maps between Riemann surfaces as covering maps, giving an introduction to the relation between covering spaces and Galois extenstions, also giving a theorem which is called "the main theorem" by Donaldson in his book *Riemann Surfaces*[Don11] and is commonly used in our discussion. Using "the main theorem" we prove the uniformization theorem.

In chapter 3 we study meromorphic functions on compact Riemann surfaces, mainly as ratio of two sections of a holomorphic line bundle. We first give a general introduction to vector bundles and line bundles, including the definitions of the degree.We also relate the language of line bundles on Riemann surfaces to the language of divisors on Riemann surfaces. Then we go to the two elegant and powerful theorems, the Serre duality and the Riemann-Roch theorem. We will also give the proof from the view point of sheaves. The we give the Abel-Jacobi theorems describing the moduli space of line bundles of a fixed degree on a Riemann surface. Finally as applications we give the Abel embedding and the Kodaira embedding.

Finally chapter 4 contains some interesting topics that cannot be contained in the previous chapters, including a brief introduction to Θ functions, the discussion about automorphisms of a Riemann surface (this is the part that attracts me to learn the theory at first), and the countability of the topology of a Riemann surface.

The main references are Prof. Lynn Heller's lecture notes during 2024 fall, *Riemann Surfaces*[Don11], *Lectures on Riemann Surfaces*[GF12], *Algebraic Curves and Riemann Surfaces*[Mir95], and *The Concept of a Riemann Surface*[WM13].

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Chapter 1

Basics about Riemann surfaces

In this chapter we discuss the basics about Riemann surfaces to prepare for the main topics in the following chapters.

1.1 Definition and first examples

Definition 1.1.1. A **Riemann surface** is given by the following data:

- A Hausdorff topological space X;
- A collection of open sets $U_\alpha \subset X$, where α ranges over some index set, which cover X;
- For each α, a homeomorphism

$$\psi_\alpha : U_\alpha \to \tilde{U}_\alpha,$$

where $\tilde{U}_\alpha$ is an open subset in $\mathbb{C}$, with the property that for all α, β the composite map $\psi_\alpha \circ \psi_\beta$ is *holomorphic* on its domain of definition.

Remark.

- In this definition we did not assume that the topology on X is second countable as we did while studying smooth manifolds. In fact this is automatically satisfied in the case of Riemann surfaces, which is known as **Radó's theorem**. We will encounter this in chapter 4.
- If we replace the word "holomorphic" by some other words, such as "smooth maps with Jacobian 1" or "rational functions", we will obtain different geometries such as sympletic geometry or birational geometry. As we will gradually see, it is essential to analyze how various geometric objects change under such transitions.

Example. *Open subsets of $\mathbb{C}$ are obviously Riemann surfaces defined by one chart id.*

Example. *The* ***projective line***

$$\mathbb{CP}^1 := \mathbb{C}^2 - 0/\mathbb{C}^* = \{[z_1 : z_2] : z_1, z_2 \in \mathbb{C}, (z_1, z_2) \neq 0\}$$

is a Riemann surface with charts

$$\psi_1 : U_1 = \{[z_1 : z_2] : z_1 \neq 0\} \to \mathbb{C}, [z_1, z_2] \mapsto z = \frac{z_2}{z_1},$$

$$\psi_2 : U_2 = \{[z_1 : z_2] : z_2 \neq 0\} \to \mathbb{C}, [z_1, z_2] \mapsto w = \frac{z_1}{z_2},$$

the transition map is $w = \frac{1}{z}$. $\mathbb{CP}^1$ *is homeomorphic to* S^2*, we leave this as an exercise to the reader.*

Example. *The* ***complex tori*** *are Riemann surfaces.*
Fix ω_1 *and* ω_2 *to be two complex numbers which are linearly independent over* $\mathbb{R}$*. Define* L *to be the lattice*

$$L = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2.$$

Then L *is a discrete additive subgroup of* $\mathbb{C}$*.Let* $X = \mathbb{C}/L$ *be the quotient group with quotient topology, the* X *is a compact connected Riemann surface whose charts are inclusions and chart domains are discs of radius* $r < \frac{1}{2}\inf_{\omega \in L} |\omega|$*, and whose transition functions are just translations by elements in* L*.*

Example. *The* graph *of holomorphic functions are obviously Riemann surfaces with the projections as charts.*

To give the next examples, we need the following theorem.
Let $P(z, w)$ be a holomorphic(i.e. holomorphic w.r.t. both variables) function on some open subset of $\mathbb{C}^2$, and let $X \subset \mathbb{C}^2$ be the zero set of P.

Theorem 1.1.2 (Implicit Function Theorem)**.** *Suppose* (z_0, w_0) *is a point in* X *and* $\frac{\partial P}{\partial w}$ *does not vanish at* (z_0, w_0)*. Then there is a disc* D_1 *centered at* z_0 *in* $\mathbb{C}$*, a disc centered at* w_0 *and a holomorphic map* $\phi : D_1 \to D_2$ *with* $\phi(z_0) = w_0$ *such that*

$$X \cap (D_1 \times D_2) = \{(z, \phi(z)) : z \in D_1\}.$$

Proof. We use the proof in [Don11].

To prove the theorem, recall that if f is a holomorphic function on an open set containing the closure of a disc D which does not vanish on the boundary ∂D, then the number of solutions to the equation $f(w) = 0$ in D, counted with multiplicity, is given by the contour integral

$$\frac{1}{2\pi i}\int_{\partial D} \frac{f'(w)}{f(w)} \mathrm{d}w$$

If there is only one solution, w_1, it is given by another contour integral,

$$w_1 = \frac{1}{2\pi i}\int_{\partial D} \frac{wf'(w)}{f(w)} \mathrm{d}w.$$

We apply these formulae to the family of functions of the variable w $f_z(w) = P(z, w)$, where we regard z as a parameter.

First, take $z = z_0$. Then the hypothesis that $\frac{\partial P}{\partial w} \neq 0$ means that f'_{z_0} does not vanish at w_0. Thus we can find a small disc D_2 centered at w_0 so that f_{z_0} has no other zeros in the closure of D_2.

Since the boundary of D_2 is compact, there is some $\delta > 0$ such that $|f_z| > 2\delta$ on ∂D_2. By continuity, this means that if z is sufficiently close to z_0, we still have $|f_z| > \delta$, say, on ∂D_2.

Thus we can apply the formula above for the number of roots of the equation $f_z(w) = 0$.

When $z = z_0$, this number must be 1. So, by continuity, the same is true for z close to z_0.

Then we define $\phi(z)$ to be this unique root. The second formula shows that

$$\phi(z) = \int_{\partial D_2} \frac{w}{P} \frac{\partial P}{\partial w} \mathrm{d}w,$$

which is clearly holomorphic in the variable z. This completes the proof. □

As a application, we see the following example.

Example. *If* $\mathrm{d}P = \frac{\partial P}{\partial z}\mathrm{d}z + \frac{\partial P}{\partial w}\mathrm{d}w$ *has no zero on* X*, then* X *is a Riemann surface.*

Similarly, we have examples in $\mathbb{CP}^2$ that are compact. For simplicity we only discuss about the case of **projective curves**.

Proposition 1.1.3. *Suppose* $p(Z_0, Z_1, Z_2)$ *is a homogeneous polynomial of degree* $d \geq 1$ *and the only solution of the equations* $\frac{\partial P}{\partial Z_1} = \frac{\partial P}{\partial Z_2} = \frac{\partial P}{\partial Z_3} = 0$ *is* $Z_1 = Z_2 = Z_3 = 0$*. Then the solutions of the equation* $p = 0$ *in* $\mathbb{CP}^2$ *form a compact Riemann surface.*

Proof. We leave this proof as an (easy) exercise to the reader. □

Next comes examples of any given genus that can be constructed now. (However probably the best way to compute the genera of these surfaces is to use the Riemann-Hurwitz formula, while the direct geometric imaginations still makes sense.)

Example. *The* ***hyper elliptic surfaces*** *are (compactifications of) surfaces of the form*

$$\{(y, z) : y^2 = p(z)\},$$

where p *is a polynomial of degree* $d \geq 5$ *without repeated roots. (The cases* $deg(p) = 3$ *or* $deg(p) = 4$ *give* ***elliptic curves****, and they are tori.)*

Here the requirement that p *has no repeated roots comes from the condition assumed in theorem 1.1.2 to ensure the regularity of the surfaces.*

Now we consider the compactification of these surfaces. Write $p(z) = \prod_{k=1}^{d}(z - z_k)$*, we have two cases depending on whether* d *is even or odd.*

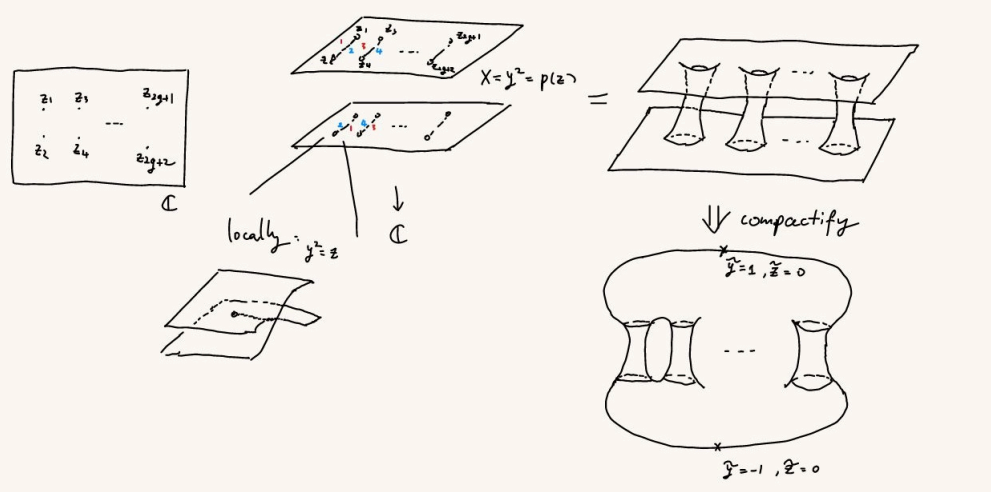


Figure 1.1: The case when $d = 2g + 2$

- *If $d = 2g + 2$, under the transition*

$$\tilde{z} = \frac{1}{z}, \tilde{y} = z^{-(g+1)}y \tag{1.1.1}$$

the relation now becomes

$$\tilde{y}^2 = \prod_{k=1}^{d}(1 - z_k\tilde{z}). \tag{1.1.2}$$

So what we actually did is to attach the original surface in $\mathbb{C}^2$ with the surface defined by 1.1.2 in $\mathbb{C}^2$ via transition 1.1.1.

In this case we actually just added two points to the original surface, which are the points where $\tilde{z} = 0$, i.e. $z = \infty$. See the figure above for what this means.

- *If $d = 2g + 1$, under the transition*

$$\tilde{z} = \frac{1}{z}, \tilde{y} = z^{-(g+1)}y \tag{1.1.3}$$

the relation now becomes

$$\tilde{y}^2 = \tilde{z}\prod_{k=1}^{d}(1 - z_k\tilde{z}). \tag{1.1.4}$$

Then what we actually did is to attach the original surface in $\mathbb{C}^2$ with the surface defined by 1.1.4 in $\mathbb{C}^2$ via transition 1.1.3.

In this case we actually just added one point to the original surface, which is the point where $\tilde{z} = 0$, i.e. $z = \infty$.

Now we turn to an elementary introduction to functions and maps on Riemann surfaces. The following theorems are rather exercises of the complex analysis course so we just give some of them and leave the others as exercises.

Definition 1.1.4. Let X be a Riemann surface and $Y \subset X$ an open subset. A function $f : Y \to \mathbb{C}$ is called **holomorphic**, if for every chart $\psi : U \to V$ on X the function

$$f \circ \psi^{-1} : \psi(U \cap Y) \to \mathbb{C}$$

is holomorphic in the usual sense.

The set of all functions holomorphic on Y is denoted by $\mathscr{O}(Y)$ or $\mathscr{O}_Y$ or even $\mathscr{O}$ if Y is clear.

Theorem 1.1.5 (Riemann's Removable Singularities Theorem)**.** *Let V be an open subset of a Riemann surface and let $a \in V$. Suppose the function $f \in \mathscr{O}(V - \{a\})$ is bounded in some neighborhood of a. Then f can be extended uniquely to a function $\tilde{f} \in \mathscr{O}(V)$.*

Definition 1.1.6. Suppose X and Y are Riemann surfaces. A continuous mapping $f : X \to Y$ is called **holomorphic**, if for every pair of charts $\psi_1 : U_1 \to V_1$ on X and $\psi_2 : U_2 \to V_2$ on Y with $f(U_1) \subset U_2$, the mapping

$$\psi_2 \circ f \circ \psi_1^{-1} : V_1 \to V_2$$

is holomorphic in the usual sense.

A mapping $f : X \to Y$ is called **biholomorphic** if it is bijective and both f and f^{-1} are holomorphic. Two Riemann surfaces X and Y are called **isomorphic** if there exists a biholomorphic mapping $f : X \to Y$.

Theorem 1.1.7 (Identity Theorem)**.** *Suppose X and Y are Riemann surfaces and $f_1, f_2 : X \to Y$ are two holomorphic mappings which coincide on a set $A \subset X$ having a limit point $a \in X$. Then f_l and f_2 are identically equal.*

Definition 1.1.8. Let X be a Riemann surface and Y be an open subset of X. By a **meromorphic function** on Y we mean a holomorphic function $f : Y' \to \mathbb{C}$, where $Y' \subset Y$ is an open subset, such that the following hold

- $Y - Y'$ contains only isolated points.
- For every point $p \in Y - Y'$ one has

$$\lim_{x \to p} |f(x)| = \infty$$

The points of $Y - Y'$ are called the **poles** of f. The set of all meromorphic functions on Y is denoted by $\mathscr{M}(Y)$.

Remark. Meromorphic functions are actually holomorphic maps to $\mathbb{CP}^1$, we leave the proof as a exercise.

Note that the Identity theorem also holds for meromorphic functions.

Theorem 1.1.9 (Local Standard Form of Holomorphic Mappings)**.** *Suppose X and Y are Riemann surfaces and $f : X \to Y$ is a non-constant holomorphic mapping. Suppose $a \in X$ and $b := f(a)$. Then there exists an integer $k \geq 1$ and charts $\varphi : U \to V$ on X and $\psi : U' \to V'$ on Y with the following properties:*

- $a \in U, \varphi(a) = 0$; $b \in U', \psi(b) = 0$.
- $f(U) \subset U'$.
- *The map $F := \psi \circ f \circ \varphi^{-1} : V \to V'$ is given by*

$$F(z) = z^k \text{ for all } z \in V.$$

Corollary 1.1.10. *Let X and Y be Riemann surfaces and let $f : X \to Y$ be an injective holomorphic mapping. Then f is a biholomorphic mapping of X onto $f(X)$.*

Proof. Apply theorem 1.1.9, noticing that $k = 1$ in this case. □

Theorem 1.1.11 (Open Mapping Theorem)**.** *Non-constant holomorphic mappings between Riemann surfaces are open mappings.*

Corollary 1.1.12 (Maximum Principle)**.** *Suppose X is a Riemann surface and $f : X \to \mathbb{C}$ is a non-constant holomorphic function. Then the absolute value of f does not attain its maximum.*

Theorem 1.1.13. *Suppose X and Y are Riemann surfaces. Suppose X is compact and $f : X \to Y$ is a non-constant holomorphic mapping. Then Y is compact and f is surjective.*

Proof. $f(X)$ is clopen in this case. □

Corollary 1.1.14. *Every holomorphic function on a compact Riemann surface is constant.*

Corollary 1.1.15. *Every meromorphic function f on $\mathbb{CP}^1$ is rational*

Theorem 1.1.16 (Liouville's Theorem)**.** *Every bounded holomorphic function $f : \mathbb{C} \to \mathbb{C}$ is constant.*

Proof. This is just a corollary of theorem 1.1.5 and theorem 1.1.13. □

1.2 Basics about sheaves

In this section we give a general introduction to presheaves and sheaves, which can be thought of as generalizations of $\mathscr{O}$, $\mathscr{M}$, or C^∞ functions.

Definition 1.2.1. Suppose X is a topological space and $\mathscr{T}$ is the system of open sets in X. A **presheaf** of abelian groups on X is a pair $(\mathscr{F}, \rho)$ consisting of

- a family $\mathscr{F} = (\mathscr{F}(U))_{U \in \mathscr{T}}$ of abelian groups,
- a family $\rho = (\rho_V^U)_{U,V \in \mathscr{T}, V \subset U}$ of group homomorphisms

$$\rho_V^U : \mathscr{F}(U) \to \mathscr{F}(V)$$

with the following properties:

$$\rho_U^U = id_{\mathscr{F}(U)} \text{ for every } U \in \mathscr{T},$$

$$\rho_W^V \circ \rho_V^U = \rho_W^U \text{ for } W \subset V \subset U.$$

Remark. Generally one just writes $\mathscr{F}$ instead of $(\mathscr{F}, \rho)$. The homomorphisms ρ_V^U are called **restriction homomorphisms**. Instead of $\rho_V^U(f)$ for $f \in \mathscr{F}(U)$ one writes just $f|_V$.

Analogous to presheaves of abelian groups one can also define presheaves of vector spaces, rings, sets, etc. The reader might have already noticed that a presheaf is nothing else but a contravariant functor from the category of open sets in X, however we will not use much category theory in this note.

The presheaf information tells us how to go to small pieces from large ones, now the sheaf axioms tells us how to reverse this, just like putting together puzzles.

Definition 1.2.2 (Sheaf Axioms)**.** A presheaf $\mathscr{F}$ on a topological space X is called a **sheaf** if for every open set $U \subset X$ and every family of open subsets $U_i \subset U, i \in I$, such that $U = \bigcup_{i \in I} U_i$ the following conditions, which we will call the Sheaf Axioms, are satisfied:

(I) If $f, g \in \mathscr{F}(U)$ are elements such that $f|_{U_i} = g|_{U_i}$ for every $i \in I$, then $f = g$.

(II) Given elements $f_i \in \mathscr{F}(U_i), iE \in I$, such that

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j} \text{ for all } i, j \in I$$

then there exists an $f \in \mathscr{F}(U)$ such that $f|_{U_i} = f_i$ for every $i \in I$.

Now we introduce some examples of sheaves. As we will see, "local properties" usually gives us sheaves.

Example. *The followings are all sheaves on Riemann surfaces.*

- $\mathscr{C}_X^\infty$ *:* $\mathscr{C}_X^\infty(U) = \{\mathscr{C}^\infty$ *functions* $f : U \to \mathbb{C}\}$.
- $\mathscr{O}_X$ *:* $\mathscr{O}_X(U) = \{$*holomorphic functions* $f : U \to \mathbb{C}\}$.

- $\mathscr{M}_X$: $\mathscr{M}_X(U) = \{meromorphic\ functions\ on\ U\}$.

These are examples of sheaves of rings (ℂ-algebras)*, below we list some sheaves of* groups (ℂ-vector spaces)*:*

- $\mathscr{O}_X^*$: $\mathscr{O}_X^*(U) = \{holomorphic\ functions\ f : U \to \mathbb{C}^*\}$. *This is the sheaf of nowhere vanishing holomorphic functions whose group operation is given by pointwise multiplication.*
- $\mathscr{M}_X^*$: $\mathscr{M}_X^*(U) = \{meromorphic\ functions\ not\ identically\ 0\ on\ any\ component\ of\ U\}$.
- $\mathscr{E}_X^1$: $\mathscr{E}_X^1(U) = \{\mathscr{C}^\infty\ 1\text{-}froms\ on\ U\}$.
- $\mathscr{E}_X^2$: $\mathscr{E}_X^2(U) = \{\mathscr{C}^\infty\ 2\text{-}froms\ on\ U\}$.

In all these examples, ρ are just natural restrictions.

Example (Locally Constant Sheaves)**.** *Let G be an arbitrary group, let*

$$\underline{G}(U) = \{locally\ constant\ functions\ f : U \to G\},$$

then this is a sheaf. The most commonly used examples are

$$\underline{\mathbb{Z}}, \quad \underline{\mathbb{R}}, \quad \underline{\mathbb{C}}.$$

These are usually called just ***constant sheaves****, while one has to remember they consist of* locally *constant functions.*

Example (Skyscraper Sheaves)**.** *Suppose that one gives, for each point $p \in X$, a different group G_p. Then to any open set U of X, one may assign the direct product of all groups G_p for $p \in U$ to U:*

$$S(U) = \prod_{p \in U} G_p.$$

One has the restriction maps given by the natural projections. It is elementary to check that this gives a sheaf on X. Since there is absolutely no requirement that the different groups at different points be related in any way, we might call such a sheaf a ***totally discontinuous sheaf****.*

An example of this is to use the same group G at every point of X. Thenone simply obtains the sheaf of all functions from X to G.

One of the most common types of totally discontinuous sheaves is obtained by giving a single group G at a single point p, and assigning the trivial group $\{0\}$ to all other points of X. This kind of sheaf is also called a ***skyscraper sheaf*** *(out of the drab skyline the group G rises majestically at the chosen point p). Such a skyscraper sheaf would be denoted by G_p. Note that this type of skyscraper sheaf has*

$$G_p(U) = \begin{cases} G & if\ p \in U, \\ 0 & if\ p \notin U. \end{cases}$$

Given a totally discontinuous sheaf S, a section $s \in S(U)$ may be "evaluated" at a point p in U by setting $s(p)$ equal to the p-th coordinate of s (which is an element of the group G, associated to the point p). The support of a section $s \in S(U)$ is the set of points $p \in U$ such that $s(p) \neq 0$. In general, a totally discontinuous sheaf can have sections with arbitrary support sets.

Notice that the sections of the skyscraper sheaf G_p can only have the point p (or the empty set) for support.

A slight variation of the G_p construction is actually more useful in many instances, and is also referred to as a ***skyscraper sheaf****. Again one gives a group G_p for every point $p \in X$, but to each open set U one assigns only the subgroup of the above direct product consisting of sections with discrete support. That is, the sections are those U-tuples $(g_p \in G, p \in U)$ such that the set of $p \in U$ with $g_p \neq 0$ forms a discrete set. This also forms a sheaf, and this case arises more often in practice.*

Here it is not so much that the sheaf itself resembles a skyscraper on the landscape, but each section of the sheaf does! Note that a totally discontinuous sheaf is a skyscraper sheaf if and only if one uses the trivial group at all but a discrete set of points of X. In particular, the skyscraper sheaf G_p is a skyscraper sheaf in this second sense.

A more complicated example is given by assigning the group of integers to every point of a Riemann surface. Then to every open set U one obtains the group of functions from U to Z, which are discretely supported. Such a function is exactly a ***divisor*** *on U. Therefore we obtain the (skyscraper)* ***sheaf of divisors*** *Div_X.*

Suppose $\mathscr{F}$ is a presheaf of sets on a topological space X and $a \in X$ is a point. On the disjoint union

$$\bigsqcup_{U \ni a} \mathscr{F}(U),$$

where the union is taken over all the open neighborhoods U of a, introduce an equivalence relation $\sim_a$ as follows: Two elements $f \in \mathscr{F}(U)$ and $g \in \mathscr{F}(V)$ are related $f \sim_a g$ precisely if there exists an open set W with $a \in W \subset U \cap V$ such that $f|_W = g|_W$. One can easily check that this really is an equivalence relation. The set $\mathscr{F}_a$ of all equivalence classes, the so-called inductive limit of $\mathscr{F}(V)$, is given by

$$\mathscr{F}_a := \varinjlim_{U \ni a} \mathscr{F}(U) = \left(\bigsqcup_{U \ni a} \mathscr{F}(U) \right) \Big/ \sim_a$$

and is called the **stalk** of $\mathscr{F}$ at the point a. If $\mathscr{F}$ is a presheaf of abelian groups (resp. vector spaces, rings), then the stalk $\mathscr{F}_a$ with the operation defined on the equivalence classes by means of the operation defined on representatives, is also an abelian group (resp. vector space, ring).

For any open neighborhood U of a, let

$$\rho_a : \mathscr{F}(U) \to \mathscr{F}_a$$

be the mapping which assigns to each element $f \in \mathscr{F}(U)$ its equivalence class modulo $\sim_a$. One calls $\rho_a(f)$ the **germ** of f at a.

Example. *Consider the sheaf $\mathscr{O}$ of holomorphic functions on a domain $X \subset \mathbb{C}$. Let $a \in X$. A germ of a holomorphic function $\varphi \in \mathscr{O}_a$ is represented by a holomorphic function in an open neighborhood of a and thus has a Taylor series expansion*

$$\sum_{v=0}^{\infty} c_v (z-a)^v$$

with a positive radius of convergence.

Two holomorphic functions on neighborhoods of a determine the same germ at a precisely if they have the same Taylor series expansion about a. Thus there is an isomorphism between the stalk $\mathscr{O}_a$ and the ring $\mathbb{C}\{z-a\}$ of all convergent power series in $z-a$ with complex coefficients.

In an analogous way, the ring $\mathscr{M}_a$ of germs of meromorphic functions at a is isomorphic to the ring of all convergent Laurent series

$$\sum_{v=k}^{\infty} c_v (z-a)^v, \quad k \in \mathbb{Z}, \quad c_v \in \mathbb{C},$$

which have finite principal parts.

For any germ of a function $\varphi \in \mathscr{O}_a$, the value of the function, $\varphi(a) \in \mathbb{C}$, is well-defined, i.e., is independent of the choice of representative.

The next lemma shows how a section of a sheaf is determined by its germs.

Lemma 1.2.3. *Suppose $\mathscr{F}$ is a sheaf of abelian groups on the topological space X and $U \subset X$ is an open subset. Then an element $f \in \mathscr{F}(U)$ is zero precisely if all germs $\rho_x(f) \in \mathscr{F}_x, x \in U$, vanish.*

Proof. This is a direct application of Sheaf Axiom I. □

Suppose X is a topological space and $\mathscr{F}$ is a presheaf on X. Let

$$|\mathscr{F}| := \bigsqcup_{x \in X} \mathscr{F}_x$$

be the disjoint union of all the stalks. Denote by

$$p : |\mathscr{F}| \to X$$

the mapping which assigns each element $\varphi \in \mathscr{F}_x$ the point x.

Now introduce the topology on $|\mathscr{F}|$ as follows: For any open subset $U \subset X$ and an element $f \in \mathscr{F}(U)$, let

$$[U, f] := \{\rho_x(f) : x \in U\} \subset |\mathscr{F}|.$$

Theorem 1.2.4. *The system $\mathfrak{B}$ of all sets $[U, f]$, where U is open in X and $f \in \mathscr{F}(U)$, is a basis for a topology on $|\mathscr{F}|$. The projection $p : |\mathscr{F}| \to X$ is a local homeomorphism.*

Proof. (a) To see that $\mathfrak{B}$ forms a basis for a topology on $|\mathscr{F}|$, one has to verify the following two conditions:

(i) Every element $\varphi \in |\mathscr{F}|$ is contained in at least one $[U, f]$. This is trivial.

(ii) If $\varphi \in [U, f] \cap [V, g]$, then there exists a $[W, h] \in \mathfrak{B}$ such that $\varphi \in [W, h] \subset [U, f] \cap [V, g]$. For suppose $p(\varphi) = x$. Then $x \in U \cap V$ and $\varphi = \rho_x(f) = \rho_x(g)$. Hence there exists an open neighborhood $W \subset U \cap V$ of x such that $f|_W = g|_W =: h$. This implies $\varphi \in [W, h] \subset [U, f] \cap [V, g]$.

(b) Now we will show that $p : |\mathscr{F}| \to X$ is a local homeomorphism.

Suppose $\varphi \in |\mathscr{F}|$ and $p(\varphi) = x$. There exists a $[U, f] \in \mathfrak{B}$ with $\varphi \in [U, f]$. Then $[U, f]$ is an open neighborhood of φ and U is an open neighborhood of x. The mapping $p : [U, f] \to U$ is bijective and also continuous and open as one sees immediately from the definition of the topology. Thus $p : |\mathscr{F}| \to X$ is a local homeomorphism.

□

Definition 1.2.5. A presheaf $\mathscr{F}$ on a topological space X is said to satisfy the **Identity Theorem** if the following holds:

If $Y \subset X$ is a domain and $f, g \in \mathscr{F}(Y)$ are elements whose germs $\rho_a(f)$ and $\rho_a(g)$ coincide at a point $a \in Y$, then $f = g$.

Example. *$\mathscr{O}$ and $\mathscr{M}$ satisfy the identity theorem.*

Theorem 1.2.6. *Suppose X is a locally connected Hausdorff space and $\mathscr{F}$ is a presheaf on X which satisfies the Identity Theorem. Then the topological space $|\mathscr{F}|$ is Hausdorff.*

Proof. Suppose $\varphi_1, \varphi_2 \in |\mathscr{F}|$ and $\varphi_1 \neq \varphi_2$. We have to find disjoint neighborhoods of φ_1 and φ_2.

(i) Suppose $p(\varphi_1) =: x \neq y := p(\varphi_2)$. Since X is Hausdorff, there exist disjoint neighborhoods U and V of x and y respectively. Then $p^{-1}(U)$ and $p^{-1}(V)$ are disjoint neighborhoods of φ_1 and φ_2 respectively.

(ii) Suppose $p(\varphi_1) = p(\varphi_2) =: x$. Suppose the germs $\varphi_i \in \mathscr{F}_x$ are represented by elements $f_i \in \mathscr{F}(U_i)$, where U_i are open neighborhoods of x, $i = 1, 2$. Let $U \subset U_1 \cap U_2$ be a connected open neighborhood of x. Then $[U, f_i|_U]$ are open neighborhoods of φ_i. Now suppose there exists $\psi \in [U, f_1|_U] \cap [U, f_2|_u]$. Let $p(\psi) = y$. Then $\psi = \rho_y(f_1) = \rho_y(f_2)$. From the Identity Theorem it follows that $f_1|_U = f_2|_U$, thus $\varphi_1 = \varphi_2$. Contradiction! Hence $[U, f_1|_U]$ and $[U, f_2|_U]$ are disjoint.

□

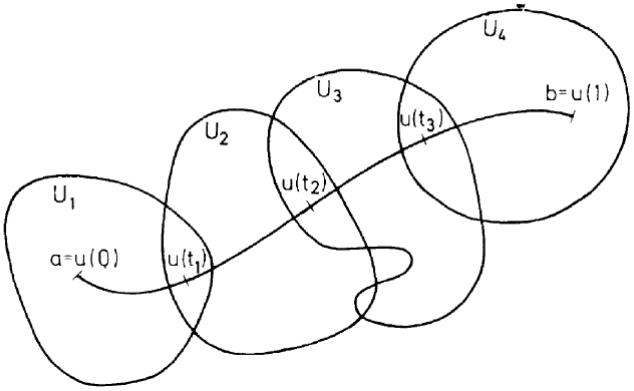


Figure 1.2: figure from [GF12] about analytic continuation along a curve

1.3 Analytic continuation

One is assumed to have basic knowledge about covering spaces while reading this section. You can read (part of) section 2.1 to get this basic understanding. In this section we follow [GF12].

Now we consider the construction of Riemann surfaces which arise from the analytic continuation of germs of functions.

Definition 1.3.1. Suppose X is a Riemann surface. $u : [0,1] \to X$ is a curve and $a := u(0), b := u(1)$. The holomorphic function germ $\psi \in \mathscr{O}_b$ is said to result from the **analytic continuation along the curve** u of the holomorphic function germ $\phi \in \mathscr{O}_a$ if the following holds:

There exists a family $\phi_t \in \mathscr{O}_{u(t)}, t \in [0,1]$ of function germs with $\phi_0 = \phi$ and $\phi_1 = \psi$ with the property that for every $t \in [0,1]$ there exists a neighborhood $T \subset [0,1]$ of t, an open set $V \subset X$ with $u(T) \subset V$ and a function $f \in \mathscr{O}(V)$ such that

$$\rho_{u(t)}(f) = \phi_t \text{ for every } t \in T.$$

Because of the compactness of $[0,1]$ this condition is equivalent to the following:

There exist a partition $0 = t_0 < t_1 < \ldots < t_{n-1} < t_n = 1$ of the interval $[0,1]$, domains $U_i \subset X$ with $u([t_{i-1}, t_i]) \subset U_i$ and holomorphic functions $f_i \in \mathscr{O}(U_i)$ for $i = 1, \ldots, n$ such that:

(i) ϕ is the germ of f_1 at the point a and ψ is the germ of f_n at the point b.

(ii) $f_i|_{V_i} = f_{i+1}|_{V_i}$ for $i = 1, \ldots, n-1$, where V_i denotes the connected component of $U_i \cap U_{i+1}$ containing the point $u(t_i)$.

If one carries out the construction given in the last section for the sheaf $\mathscr{O}$ of holomorphic functions, then one gets a map $p : |\mathscr{O}| \to X$. The next lemma shows that one can interpret analytic continuation along a curve by means of this map.

Lemma 1.3.2. *Suppose X is a Riemann surface and $u : [0,1] \to X$ is a curve in X with $u(0) =: a$ and $u(1) =: b$. Then a function germ $\psi \in \mathscr{O}_b$ is the analytic continuation of a function germ $\phi \in \mathscr{O}_a$ along u precisely if there exists a lifting $\hat{u} : [0,1] \to |\mathscr{O}|$ of the curve u such that $\hat{u}(0) = \phi$ and $\hat{u}(1) = \psi$.*

Proof.

- Suppose $\psi \in \mathscr{O}_b$ is the analytic continuation of $\phi \in \mathscr{O}_a$ along u. Let $\phi_t \in \mathscr{O}_{u(t)}, t \in [0,1]$ be the family of function germs as given in the definition. It follows directly from the definition of the topology of $|\mathscr{O}|$ that the correspondence $t \mapsto \phi_t$ represents a continuous mapping $\hat{u} : [0,1] \to |\mathscr{O}|$. Thus $\hat{u}$ is a lifting of u with $\hat{u}(0) = \phi$ and $\hat{u}(1) = \psi$.

- Suppose there is a lifting $\hat{u} : [0,1] \to |\mathscr{O}|$ of u with $\hat{u}(0) = \phi$ and $\hat{u}(1) = \psi$. For $t \in [0,1]$, let $\phi_t := \hat{u}(t)$. Then $\hat{u}(0) = \phi$ and $\hat{u}(1) = \psi$. Let $\tau \in [0,1]$ and suppose $[U, f] \subset |\mathscr{O}|$ is an open neighborhood of $u(\tau)$. Then there exists a neighborhood $T \subset [0,1]$ of τ such that $u(T) \subset [U, f]$. This implies $u(T) \subset U$ and $\phi_t = \hat{u}(t) = \rho_{u(t)}(f)$ for every $t \in T$. But this means that ψ is the analytic continuation of ϕ along u.

□

Because of the uniqueness of liftings (theorem 2.1.5 and theorem 1.2.6) it follows from the lemma that if the analytic continuation of a function germ exists, then it is uniquely determined. Another consequence of the lemma is the Monodromy Theorem.

Theorem 1.3.3 (Monodromy Theorem)**.** *Suppose X is a Riemann surface and $u_0, u_1 : [0,1] \to X$ are homotopic curves from a to b. Suppose $u_s, 0 \leq s \leq 1$, is a deformation of u_0 into u_1 and $\phi \in \mathscr{O}_a$ is a function germ which admits an analytic continuation along every curve u_s. Then the analytic continuations of ϕ along u_0 and u_1 yield the same function germ $\psi \in \mathscr{O}_b$.*

Proof. Apply the fact that liftings of homotopic curves have the same end point (under certain conditions, which are satisfied by $p : |\mathscr{O}| \to X$, see theorem 2.1.8) to the curves u_s and $p : |\mathscr{O}| \to X$, we obtain that $\hat{u}_0(1) = \hat{u}_1(1) = \psi$. □

Corollary 1.3.4. *Suppose X is a* simply connected *Riemann surface, $a \in X$ and $\phi \in \mathscr{O}_a$ is a function germ which admits an analytic continuation along every curve starting at a. Then there exists a globally defined holomorphic function $f \in \mathscr{O}(X)$ such that $\rho_a(f) = \phi$.*

Proof. This is automatic using theorem 1.3.3. □

In general, even if the analytic continuation of some function germ is possible along two curves with the same initial and end points, then the resulting germs at the end point may be different. Thus if we consider all the germs arising by analytic continuation from the given function germ we get a multi-valued function. Our next task is to look at this situation and to make the details precise.

Suppose X and Y are Riemann surfaces and $\mathscr{O}_X$ and $\mathscr{O}_Y$ are the sheaves of holomorphic functions on them. Suppose $p : Y \to X$ is an unbranched holomorphic map. Since p is locally biholomorphic, for each $y \in Y$ it induces an isomorphism

$$p^* : \mathscr{O}_{X,p(y)} \to \mathscr{O}_{Y,y}.$$

Let

$$p_* : \mathscr{O}_{Y,y} \to \mathscr{O}_{X,p(y)}$$

be the inverse of p^*.

Definition 1.3.5. Suppose X is a Riemann surface, $a \in X$ is a point and $\phi \in \mathscr{O}_a$ is a function germ. A quadrupel (Y, p, f, b) is called an **analytic continuation** of ϕ if:

(i) Y is a Riemann surface and $p : Y \to X$ is an *unbranched* holomorphic map.

(ii) f is a holomorphic function on Y.

(iii) b is a point of Y such that $p(b) = a$ and

$$p_*(\rho_b(f)) = \phi.$$

An analytic continuation (Y, p, f, b) of ϕ is said to be **maximal** if it has the following universal property:

If (Z, q, g, c) is any other analytic continuation of ϕ, then there exists a fiber-preserving holomorphic mapping $F : Z \to Y$ such that $F(c) = b$ and $F^*(f) := f \circ F = g$.

Remark. It is an exercise to prove that the maximal analytic continuation is (if exists,), up to isomorphism, unique.

Lemma 1.3.6. *Suppose X is a Riemann surface, $a \in X$, $\phi \in \mathscr{O}_a$ and (Y, p, f, b) is an analytic continuation of ϕ. Then if $v : [0, 1] \to Y$ is a curve with $v(0) = b$ and $v(1) =: y$, then the function germ $\psi := p_*(\rho_y(f)) \in \mathscr{O}_{p(y))}$ is an analytic continuation of ϕ along the curve $u := p \circ v$.*

Proof. Exercise, for a detailed proof, see [GF12]. □

Theorem 1.3.7 (Existence of the Maximal Analytic Continuation)**.** *Suppose X is a Riemann surface, $a \in X$ and $\phi \in \mathscr{O}_a$ is a holomorphic function germ at the point a. Then there exists a maximal analytic continuation (Y, p, f, b) of ϕ.*

Proof. Let Y be the connected component of $|\mathscr{O}|$ containing ϕ. Let p also denote the restriction of the mapping $p : |\mathscr{O}| \to X$ to Y. Then $p : Y \to X$ is a local homeomorphism. By theorem 2.1.4 there is a complex structure on Y so that it becomes a Riemann surface and the mapping $p : Y \to X$ is holomorphic.

Now define a holomorphic function $f : Y \to \mathbb{C}$ as follows. By definition every $\eta \in Y$ is a function germ at the point $p(\eta)$. Set $f(\eta) := \eta(p(\eta))$. One easily sees that f is

holomorphic and $p_*(\rho_\eta(f)) = \eta$ for every $\eta \in Y$. Thus if one lets $b := \phi$, then (Y, p, f, b) is an analytic continuation of ϕ.

Now we will show that (Y, p, f, b) is a maximal analytic continuation of ϕ.

Suppose (Z, q, g, c) is another analytic continuation of ϕ. Define the map $F : Z \to Y$ as follows.

Suppose $\zeta \in Z$ and $q(\zeta) =: x$. By the lemma above the function germ $q_*(\rho_\zeta(g)) \in \mathscr{O}_x$ arises by analytic continuation along a curve from a to x from the function germ ϕ. By lemma above Y consists of all function germs which are obtained by the analytic continuation of ϕ along curves. Hence there exists exactly one $\eta \in Y$ such that $q_*(\rho_\zeta(g)) = \eta$. Let $F(\zeta) = \eta$. It is easy to check that $F : Z \to Y$ is a fiber-preserving holomorphic map (a lifting of $q : Z \to X$) such that $F(c) = b$ and $F^*(f) = g$. □

Remark. The analytic continuation of meromorphic function germs can be handled by using the techniques employed in this section for holomorphic function germs. One just looks at the map $|\mathscr{M}|$.

1.4 Calculus

In this section we follow [Don11].

Let X be a Riemann surface and so, a smooth oriented surface. Thus, for each point p in X, we have a tangent space TX_p-a two-dimensional real vector space. We also have a cotangent space

$$T_p^*X = \mathrm{Hom}_{\mathbb{R}}(T_pX, \mathbb{R}),$$

such that the derivative of any real-valued function on X yields an element of T^*X_p. We may just as well consider the complex cotangent space

$$T_p^*X^{\mathbb{C}} = T_p^*X \otimes_{\mathbb{R}} \mathbb{C},$$

such that the derivative of any complex-valued function on X yields an element of $T_p^*X^{\mathbb{C}}$.

By a **complex structure** on a real vector space V we mean an R-linear map $J : V \to V$ with $J^2 = -id$. We say that an $\mathbb{R}$-linear map A from V to $\mathbb{C}$ is **complex linear** if $A(Jv) = iAv$ for all v, and **complex antilinear** if $A(Jv) = -iA(v)$ for all v.

Lemma 1.4.1. *There is a unique way to define a complex structure on T_pX such that the derivative of any holomorphic function, defined on a neighbourhood of p in X, is complex linear.*

This follows immediately from the definition of a holomorphic function.

Lemma 1.4.2. *Any $\mathbb{R}$-linear map from V to $\mathbb{C}$ can be written in a unique way as a sum of complex linear and antilinear maps.*

Proof. For the existence, we write $A = A' + A''$, where

$$A'(v) = \frac{1}{2}(Av - iA(Jv)), A''(v) = \frac{1}{2}(Av + iA(Jv)),$$

and check that A' and A'' are complex linear and antilinear, respectively.

Uniqueness is similarly easy. □

Putting this together, we see that we can write the complex cotangent space as a direct sum

$$T_p^* X^{\mathbb{C}} = T_p^* X' \oplus T_p^* X''$$

in such a way that if f is a local holomorphic function, then the derivative of f lies in $T_p^* X'$ and the derivative of $\bar{f}$ lies in $T_p^* X''$. Now we can decompose the complex 1-forms on X into corresponding pieces

$$\Omega^1_{X,\mathbb{C}} = \Omega^{1,0}_X \oplus \Omega^{0,1}_X,$$

where $\Omega^1_{X,\mathbb{C}} = \mathscr{E}^1_X$ is the space of all complex valued differential forms, and elements of $\Omega^{1,0}_X$ lie in $T_p^* X'$ and elements of $\Omega^{0,1}_X$ lie in $T_p^* X''$.

We now decompose the exterior derivative operators according to this decomposition of the forms, so we get a diagram

$$\begin{array}{ccc} \Omega^{0,1} & \xrightarrow{\partial} & \Omega^2 \\ \uparrow{\scriptstyle \bar{\partial}} & & \uparrow{\scriptstyle \bar{\partial}} \\ \Omega^0 & \xrightarrow[\partial]{} & \Omega^{1,0} \end{array}$$

Let us see this more explicitly, in a complex local coordinate $z = x + iy$. Thus x, y are real coordinates as considered in the previous section. We have

$$\mathrm{d}z = \mathrm{d}x + i\,\mathrm{d}y\,, \mathrm{d}\bar{z} = \mathrm{d}x - i\,\mathrm{d}y\,,$$

and these form basis elements for T^*X', T^*X'', respectively. So a $(1,0)$-form is expressed locally as $\alpha\,\mathrm{d}z$ and a $(0,1)$-form as $\beta\,\mathrm{d}\bar{z}$ for functions α, β. If f is a complex-valued function, then

$$\mathrm{d}f = \frac{\partial f}{\partial x}\mathrm{d}x + \frac{\partial f}{\partial y}\mathrm{d}y.$$

We write

$$\mathrm{d}x = \frac{1}{2}(\mathrm{d}z + \mathrm{d}\bar{z}), \mathrm{d}y = \frac{1}{2i}(\mathrm{d}z - \mathrm{d}\bar{z}),$$

so,

$$\mathrm{d}f = \frac{1}{2}(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y})\mathrm{d}z + \frac{1}{2}(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y})\mathrm{d}\bar{z}.$$

This means that

$$\partial f = \frac{\partial f}{\partial z}\mathrm{d}z, \bar{\partial} f = \frac{\partial f}{\partial \bar{z}}\mathrm{d}\bar{z},$$

where we define

$$\frac{\partial f}{\partial z} = \frac{1}{2}(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}), \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}).$$

The equation $\bar{\partial} f = 0$ is, in this local co-ordinate, just the Cauchy–Riemann equation, as of course it should be, since by definition a function is holomorphic if and only if $\bar{\partial} f = 0$. If f is a holomorphic function, then we have, in local coordinates,

$$\mathrm{d}f = \partial f = f'(z)\mathrm{d}z,$$

where f' denotes the usual derivative of complex analysis.

Now consider the operators $\partial, \bar{\partial}$ on $\Omega^{0,1}$ and $\Omega^{1,0}$. By following through the definitions we find that, in our local co-ordinate,

$$\partial(A\mathrm{d}\bar{z}) = \frac{\partial A}{\partial z}\mathrm{d}z \wedge \mathrm{d}\bar{z} = 2i\frac{\partial A}{\partial z}\mathrm{d}x \wedge \mathrm{d}y,$$
$$\bar{\partial}(B\mathrm{d}z) = \frac{\partial B}{\partial \bar{z}}\mathrm{d}\bar{z} \wedge \mathrm{d}z = -2i\frac{\partial B}{\partial \bar{z}}\mathrm{d}x \wedge \mathrm{d}y.$$

We note here that, different form the operators $\frac{\partial}{\partial \bar{z}}$ and $\frac{\partial}{\partial z}$, we have

$$\partial\bar{\partial} = -\bar{\partial}\partial$$

since $\mathrm{d}^2 = 0$.

Definition 1.4.3. A $(1,0)$-form β is a **holomorphic 1-form** if $\bar{\partial}\beta = 0$.

Thus, in local coordinates, a holomorphic 1-form has the shape $B\mathrm{d}z$, where B is a holomorphic function.

Suppose $S \subset X$is a compact surface with boundary and α is a holomorphic 1-form on a neighborhood of S. Then α is, in particular, a closed 1-form, so Stokes' Theorem gives

$$\int_{\partial S} \alpha = 0.$$

This is one version of **Cauchy's Theorem** on a Riemann surface.

We define a **meromorphic 1-form** α on X in the obvious way: a holomorphic 1-form on $X - D$, where D is a discrete subset of X, which can be written locally as $f(z)\mathrm{d}z$, where f is a meromorphic function. (Of course, one needs to check that this is independent of the choice of local chart.) The points of the minimal such set D are the **poles** of the meromorphic 1-form. Let p be such a pole and let C be a small loop in X encircling p. We define the **residue** of α at p to be

$$\mathrm{Res}_p(\alpha) := \frac{1}{2\pi i}\int_C \alpha.$$

Clearly, this is the same as writing, in a local co-ordinate z centred at p,

$$\alpha = f(z)\mathrm{d}z,$$

where $f(z) = \sum_{j=-k}^{\infty} a_j z^j$ is a meromorphic function, and taking the usual residue a_{-1} of f ; one can just as well take this as the definition.

Proposition 1.4.4. *Suppose α is a meromorphic 1-form on a compact Riemann surface X. Then the sum of the residues of α, running over all the poles, is zero.*

On a Riemann surface, we have a natural second-order differential operator. We define

$$\Delta = 2i\bar{\partial}\partial : \Omega^0 \to \Omega^2.$$

Then, in local co-ordinates,

$$\Delta f = -(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})f.$$

Thus if, in a given local coordinate system, we identify the 2-forms with functions using the area form $\mathrm{d}x \wedge \mathrm{d}y$, the operator Δ becomes the standard Laplace operator (up to a sign convention). A function f satisfying the differential equation $\Delta f = 0$ is called a **harmonic** function. If f is a holomorphic function, then the real and imaginary parts of f are harmonic.

Conversely, we have the following lemma.

Lemma 1.4.5. *If ϕ is a real-valued harmonic function on a neighbourhood N of a point p in a Riemann surface X, then there is an open neighbourhood $U \subset N$ of p and a holomorphic function f on U with $\phi = \mathrm{Re}(f)$.*

Being local, this is not really different from the corresponding result for functions on open sets in $\mathbb{C}$, which the reader has very likely encountered in a standard complex analysis course. However, it may be helpful to see how the proof works in our notation.

Let A be the real 1-form $i\bar{\partial}\phi + \overline{i\bar{\partial}\phi}$. Then the hypothesis that $\partial\bar{\partial}\phi = 0$ shows that $\mathrm{d}A = 0$. Thus if U is a small disc about p (or any open set with $H^1(U) = 0$), we can find a real valued function ψ with $A = \mathrm{d}\psi$. This means that $\partial\psi = -i\partial\phi$ and $\bar{\partial}\psi = i\bar{\partial}\phi$. Then

$$\bar{\partial}(\phi + i\psi) = \bar{\partial}\phi + i\bar{\partial}\psi = 0.$$

So we can take $f = \phi + i\psi$.

Proposition 1.4.6 (Maximum Principle for Harmonic Functions)**.** *Suppose ϕ is a non-constant real valued harmonic function on a connected open set U in a Riemann surface X. Then, for a point x in U, there is a point x' in U with $\phi(x') > \phi(x)$.*

This can be seen by writing ϕ, near to x, as the real part of a holomorphic function and then using the open mapping theorem.

Next we discuss about the Dirichlet norm.

Let X be a Riemann surface and let α be a $(1,0)$-form on X. We consider the 2-form $i\alpha \wedge \bar{\alpha}$. In a local complex coordinate $z = x + iy$, if $\alpha = p\mathrm{d}z$ then

$$i\alpha \wedge \bar{\alpha} = i|p|^2\,\mathrm{d}z \wedge \mathrm{d}\bar{z} = 2|p|^2\,\mathrm{d}x \wedge \mathrm{d}y\,.$$

So $i\alpha \wedge \bar{\alpha}$ is a positive 2-form. We define

$$\|\alpha\|^2 = \int_X i\alpha \wedge \bar{\alpha}, \tag{1.4.1}$$

taking values in $[0, \infty]$. Of course, if α has compact support, the integral is finite and defines a norm on the space of compactly supported $(1,0)$-forms. This norm (on the compactly supported forms) is derived from a Hermitian inner product

$$\langle \alpha, \beta \rangle := \int_X i\alpha \wedge \bar{\beta}.$$

If we have an area form ω on X, we can define a pointwise norm, the function characterized by

$$i\alpha \wedge \bar{\alpha} = |\alpha|^2 \omega.$$

Then we can tautologically write,

$$\|\alpha\|^2 = \int_X |\alpha|^2 \omega.$$

This is perhaps a more familiar point of view. However, the key point is that the "L^2-norm" on $(1,0)$-forms is actually independent of the choice of an area form.

We can identify the real 1-forms with the $(1,0)$-forms by mapping a real 1-form A to its $(1,0)$ component $A^{1,0}$. Thus we define

$$\|A\|^2 := 2\left\|A^{1,0}\right\|^2.$$

Again this norm is associated to a (real-valued) inner product $\langle , \rangle$ on the compactly supported real 1-forms,

$$\langle A, B \rangle := \int_X 2iA^{1,0} \wedge B^{0,1}.$$

Lemma 1.4.7. *Let A, B be real 1-forms on a Riemann surface X. Then*

$$\int_X |A \wedge B| \leq \|A\| \|B\|.$$

Proof. Suppose first that A and B are supported inside some chart domain C. Thus we can write $A = P\mathrm{d}z + \bar{P}\,\mathrm{d}\bar{z}$, $B = Q\mathrm{d}z + \bar{Q}\,\mathrm{d}\bar{z}$ for complex-valued functions P, Q. Then

$$A \wedge B = (P\bar{Q} - \bar{P}Q)\mathrm{d}z \wedge \mathrm{d}\bar{z} = -4\,\mathrm{Im}(P\bar{Q})\mathrm{d}x \wedge \mathrm{d}y.$$

Then

$$\int_X |A \wedge B| = 4\int_C \left|\mathrm{Im}(P\bar{Q})\right| \mathrm{d}x \wedge \mathrm{d}y \leq \left(2\int_X |P|^2 \mathrm{d}x \wedge \mathrm{d}y\right)^{\frac{1}{2}} \left(2\int_X |Q|^2 \mathrm{d}x \wedge \mathrm{d}y\right)^{\frac{1}{2}} = \|A\|\|B\|,$$

by Cauchy-Schwartz inequality.

For the general case one just use a partition of unity. □

Suppose f and g are real-valued functions, at least one of compact support on X. We define the **Dirichlet inner product** to be

$$\langle f, g\rangle_D := \langle \mathrm{d}f, \mathrm{d}g\rangle = 2i\int_X \partial f \wedge \bar{\partial} g.$$

Notice that by Stokes' theorem, if f and g are supported on a chart domain C, we have

$$\langle f, g\rangle_D = 2i\int_X \partial f \wedge \bar{\partial} g = -\int_X \left(\frac{\partial f}{\partial x}\frac{\partial g}{\partial x} + \frac{\partial f}{\partial y}\frac{\partial g}{\partial y}\right) \mathrm{d}x \wedge \mathrm{d}y.$$

Likewise, we define the **Dirichlet norm** by

$$\|f\|_D = \|\mathrm{d}f\|.$$

with our usual convention that this could be $+\infty$. (Our language here ignores the fact that this is only a semi-norm, since the constants have derivative zero.)

Lemma 1.4.8. *If at least one of f , g have* compact support*, then*

$$\langle f, g\rangle_D = -\int_X f\Delta g = -\int_X g\Delta f.$$

Proof.

$$\langle f, g\rangle_D = \langle \mathrm{d}f, \mathrm{d}g\rangle = 2i\int_X \partial f \wedge \bar{\partial} g = 2i\int_X \partial(f\bar{\partial} g) - f\partial\bar{\partial} g = -\int_X f\Delta g$$

□

Chapter 2

Covering spaces

2.1 Covering spaces and branched covering spaces

In this section we just follow [GF12]. We skip some of the proofs, so anyone wants to see the details can see [GF12].

Definition 2.1.1. Suppose X and Y are topological spaces and $p : Y \to X$ is a continuous map. For $x \in X$, the set $p^{-1}(x)$ is called the **fiber** of p over x. If $y \in p^{-1}(X)$, then one says that the point y **lies over** x. If $p : Y \to X$ and $q : Z \to X$ are continuous maps, then a map $f : Y \to Z$ is called **fiber-preserving** if $p = q \circ f$.

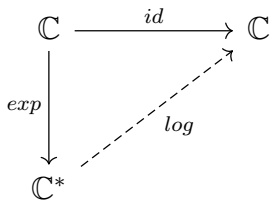
A mapping $p : Y \to X$, between topological spaces X and Y, is said to be **discrete** if the fiber $p^{-1}(x)$ of every point $x \in X$ is a discrete subset of Y.

Theorem 2.1.2. *Suppose X and Y are Riemann surfaces and $p : Y \to X$ is a non-constant holomorphic map. Then p is open and discrete.*

If $p : Y \to X$ is a non-constant holomorphic map, then we will say that Y is a **domain over** X.

A holomorphic (resp. meromorphic) function $f : Y \to \mathbb{C}$ (resp.$f : Y \to \mathbb{CP}^1$) may also be considered as a multi-valued holomorphic (meromorphic) function on X. If $x \in X$ and $p^{-1}(x) = y_j : j \in J$, then the $f(y_j), j \in J$, are the different values of this multi-valued function at the point x. Of course it might turn out that $p^{-1}(x)$ is a single point or is empty.

The following diagram shows an example (the logarithm).



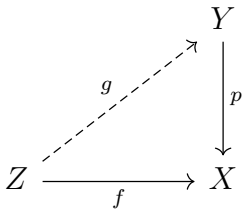
Definition 2.1.3. Suppose X and Y are Riemann surfaces and $p : Y \to X$ is a non-constant holomorphic map. A point $y \in Y$ is called a **branch point** or **ramification**

point of p, if there is no neighborhood V of y such that $p|_V$ is injective. The map p is called an **unbranched** holomorphic map if it has no branch points.

Obviously a holomorphic map between Riemann surfaces is umbranched iff it is a local homeomorphism.

Theorem 2.1.4. *Suppose X is a Riemann surface, Y is a Hausdorff topological space and $p : Y \to X$ is a local homeomorphism. Then there is a unique Riemann surface structure on Y such that p is holomorphic.*

Suppose X, Y and Z are topological spaces and $p : Y \to X$ and $f : Z \to X$ are continuous maps. Then by a lifting of f w.r.t. p is meant a continuous mapping $g : Z \to Y$ such that $f = p \circ g$, i.e. the following diagram commutes.



Theorem 2.1.5 (Uniqueness of Lifting)**.** *Suppose X and Y are Hausdorff spaces and $p : Y \to X$ is a local homeomorphism. Suppose Z is a connected topological space and $f : Z \to X$ is a continuous mapping. If $g_1, g_2 : Z \to Y$ are two liftings of f and $g_1(z_0) = g_2(z_0)$ for some point $z_0 \in Z$ then $g_1 = g_2$.*

Proof. The set $\{z \in Z : g_1(z) = g_2(z)\}$ is clopen. □

Theorem 2.1.6. *Suppose X, Y and Z are Riemann surfaces, $p : Y \to X$ is an* unbranched *holomorphic map and $f : Z \to X$ is any holomorphic map. Then every lifting $g : Z \to Y$ of f is holomorphic.*

Corollary 2.1.7. *Suppose X, Y and Z are Riemann surfaces and $p : Y \to X$ and $q : Z \to X$ are unbranched holomorphic maps. Then every continuous fiber-preserving map $f : Y \to Z$ is holomorphic.*

In the following we let $I := [0, 1]$.

Theorem 2.1.8 (Lifting of Homotopic Curves)**.** *Suppose X and Y are Hausdorff spaces and $p : Y \to X$ is a local homeomorphism. Suppose $a, b \in X$ and $\hat{a} \in Y$ is a point such that $p(\hat{a}) = a$. Further suppose a continuous mapping $A : I \times I \to X$ is given such that $A(0, s) = a$ and $A(1, s) = b$ for every $s \in I$. Set $u_s(t) := A(t, s)$.*

If every curve u_s can be lifted to a curve $\hat{u}_s$ with initial point $\hat{a}$, then $\hat{u}_0$ and $\hat{u}_1$ have the same end point and are homotopic.

Proof. See [GF12]. □

Definition 2.1.9. Suppose X and Y are topological spaces. A mapping $p : Y \to X$ is called a **covering map** if the following holds.

Every point $x \in X$ has an open neighborhood U such that its preimage $p^{-1}(U)$ can be represented as

$$p^{-1}(U) = \bigcup_{j \in J} V_j,$$

where the $V_j, j \in J$, are disjoint open subsets of Y, and all the mappings $p|_{V_j} \to U$ are homeomorphisms. In particular, p is a local homeomorphism. In this case Y is called a **covering space** over X.

Definition 2.1.10. A continuous map $p : Y \to X$ is said to have the **curve lifting property** if the following condition holds:

For every curve $u : [0, 1] \to X$ and every point $y_0 \in Y$ with $p(y_0) = u(0)$ there exists a lifting $\hat{u} : [0, 1] \to Y$ of u such that $\hat{u}(0) = y_0$.

Theorem 2.1.11. *Every covering map $p : Y \to X$ of topological spaces X and Y has the curve lifting property.*

Theorem 2.1.12. *Suppose X and Y are Hausdorff spaces with X pathwise connected and $p : Y \to X$ is a covering map. Then for any two points $x_0, x_l \in X$ the sets $p^{-1}(x_0)$ and $p^{-1}(x_1)$ have the same cardinality. In particular, if Y is non-empty, then p is surjective.*

Proof. Using lifting of paths we can construct the one-one correspondence. □

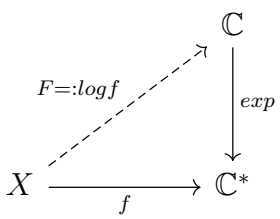
Theorem 2.1.13 (Liftting of Maps)**.** *Suppose X and Y are Hausdorff spaces and $p : Y \to X$ is a local homeomorphism with the curve lifting property. Further, suppose Z is a simply connected, pathwise connected and locally pathwise connected topological space and $f : Z \to X$ is a continuous mapping. Then for every choice of points $z_0 \in Z$ and $y_0 \in Y$ with $f(z_0) = p(y_0)$ there exists precisely one lifting $\hat{f} : Z \to Y$ such that $\hat{f}(z_0) = y_0$.*

Remark. Actually we can remove the assumption for Z to be simply connected and add

$$f_*(\pi_1(Z)) \subset p_*(\pi_1(Y))$$

instead, which is obviously also necessary.

Example. *The above theorem applies to the condition below:*



where X is simply connected and f is an arbitrary continuous map.

When f is holomorphic, by theorem 2.1.6 F is also homomorphic.

Theorem 2.1.14. *Suppose X is a manifold, Y is a Hausdorff space and $p : Y \to X$ is a local homeomorphism with the curve lifting property. Then p is a covering map.*

Proof. Suppose $x_0 \in X$ is an arbitrary point and $y_j, j \in J$, are the preimages of x_0 with respect to p. Take U to be an open neighborhood of x_0 which is homeomorphic to a ball and let $f : U \to X$ be the canonical injection. From theorem 2.1.13 it follows that for every $j \in J$ there is a lifting $\hat{f}_j : U \to Y$ of f such that $\hat{f}_j(x_0) = y_j$. Let $V_j = \hat{f}_j(U)$. Now one can easily convince oneself that

$$p^{-1}(U) = \bigcup_{j \in J} V_j,$$

that the V_j are pairwise disjoint (by the uniqueness of lifting) open sets and that every mapping $p|_{V_j} \to U$ is a homeomorphism. □

Recall that a locally compact topological space is a Hausdorff space such that every point has a compact neighborhood. A continuous mapping $f : X \to Y$ between two locally compact spaces is called **proper** if the preimage of every compact set is compact.For example this is always so if X is compact.

A proper mapping is closed, i.e., the image of every closed set is closed. This follows from the fact that in a locally compact space a subset is closed precisely if its intersection with every compact set is compact.

Lemma 2.1.15. *Suppose X and Y are locally compact spaces and $p : Y \to X$ is a proper, discrete map. Then the following hold:*

- *For every point $x \in X$ the set $p^{-l}(X)$ is finite.*
- *If $x \in X$ and V is a neighborhood of $p^{-1}(x)$, then there exists a neighborhood U of x with $p^{-l}(U) \subset V$.*

Theorem 2.1.16. *Suppose X and Y are locally compact spaces and $p : Y \to X$ is a proper local homeomorphism. Then p is a covering map.*

Suppose X and Y are Riemann surfaces and $f : X \to Y$ is a proper, non-constant, holomorphic mapping. It follows from theorem 1.1.9 that the set A of branch points of f is closed and discrete. Since f is proper, $B := f(A)$ is also closed and discrete. One calls B the set of **critical values** of f.

Let $Y' := Y - B$ and $X' := X - f^{-1}(B) \subset X - A$. Then $f : X' \to Y'$ is a proper unbranched holomorphic covering and by 2.1.16, 2.1.12 and 2.1.15 it has a well-defined finite number of sheets n. This means that every value $c \in Y'$ is taken exactly n times. In order to be able to extend this statement to the critical values $b \in B$ as well, we have to consider the multiplicities.

For $x \in X$ denote by $v(f,x)$ the multiplicity, in the sense of 1.1.9, with which f takes the values $f(x)$ at the point x. Then we will say that f takes the value $c \in Y$, counting multiplicities, m times on X, if

$$m = \sum_{x \in f^{-1}(c)} v(f,x).$$

Theorem 2.1.17 (The degree of a holomorphic map)**.** *Suppose X and Y are Riemann surfaces and $f : X \to Y$ is a proper non-constant holomorphic map. Then there exists a natural number n such that f takes every value $c \in Y$, counting multiplicities, n times. This n is called the* ***degree*** *of f.*

Proof. Using the same notation as above, let n be the number of sheets of the unbranched covering $f : X' \to Y'$. Suppose $b \in B$ is a critical value, $f^{-1}(b) = \{x_1, \ldots, x_r\}$ and $k_j := v(f, x_j)$.

By 1.1.9 there exist disjoint neighborhoods U_j of x_j and V_j of b such that for every $c \in V_j - \{b\}$ the set $f^{-1}(c) \cap U_j$ consists of exactly k_j points $(j = 1, ..., r)$.

By Lemma 2.1.15 we can find a neighborhood $V \subset V_1 \cap \cdots \cap V_r$ of b such that $f^{-1}(V) \subset U_1 \cup \cdots \cup U_r$. Then for every point $c \in V \cup Y'$ we have that $f^{-1}(c)$ consists of $k_1 + ... + k_r$ points.

On the other hand, for $c \in Y'$ the cardinality of $p^{-1}(c)$ is equal to n. Thus $n = k_1 + ... + k_r$. □

Remark. A proper non-constant holomorphic map will be called an **n-sheeted holomorphic covering map**, where n is the degree of the map. Note that holomorphic covering maps are allowed to have branch points. If we wish to emphasize that there are none, then we will specifically say that the map is *unbranched*.

If we speak of a topological covering map or if there is no complex structure, then we mean a covering map in the sense of definition 2.1.9.

Among all the covering spaces of a manifold X, there is one which deserves to be called the "largest", namely, the *universal covering*. All other covering spaces can be obtained from this one as quotients, and what happens to the universal covering when it is acted on by the group of "covering transformations" is closely related to $\pi_1(X)$.

Definition 2.1.18. Suppose X and Y are connected topological spaces and $p : Y \to X$ is a covering map. $p : Y \to X$ is called the **universal covering** of X if it satisfies the following *universal property*:

For every covering map $q : Z \to X$, with Z connected, and every choice of points $y_0 \in Y, z_0 \in Z$ with $p(y_0) = q(z_0)$ there exists exactly one continuous fiber-preserving mapping $f : Y \to Z$ such that $f(y_0) = z_0$.

Thanks to the uniqueness of liftings, we know that the universal covering is unique up to isomorphisms. It is an exercise to figure out what this means and to prove it.

Theorem 2.1.19 (Characterization of the Universal Covering by π_1)**.** *Suppose X and Y are connected manifolds, Y is simply connected and $p : Y \to X$ is a covering map. Then p is the universal covering of X.*

Theorem 2.1.20. *Suppose X is a connected manifold. Then there exists a connected, simply connected manifold $\tilde{X}$ and a covering map $p : \tilde{X} \to X$.*

Proof. See [GF12]. □

Definition 2.1.21. Suppose X and Y are topological spaces and $p : Y \to X$ is a covering map. By a **covering transformation** or **deck transformation** of this covering we mean a fiber-preserving homeomorphism $f : Y \to Y$.

With operation the composition of mappings, the set of all covering transformation of $p : Y \to X$ forms a group which we denote by $Deck(Y/X)$. If there is any chance of confusion, then we will write $Deck(Y \xrightarrow{p} X)$ instead.

Definition 2.1.22. Suppose X and Y are connected Hausdorff spaces and $p : Y \to X$ is a covering map. The covering is called **Galois** (the terms **normal** and **regular** are also in common usage) if for every pair of points $y_0, y_1 \in Y$ with $p(y_0) = p(y_l)$ there exists a covering transformation $f : Y \to Y$ such that $f(y_0) = y_1$.

There is a connection between Galois coverings and Galois field extensions, which will be discussed in the next section.

Theorem 2.1.23. *Suppose X is a connected manifold and $p : X \to \tilde{X}$ is its universal covering. Then p is Galois and $Deck(\tilde{X}/X)$ is isomorphic to the fundamental group $\pi_1(X)$.*

Proof. It's easy to see from the universal property of the universal covering that it is Galois.

Now we prove the second statement. Suppose $x_0 \in X$ and $y_0 \in Y$ are points with $p(y_0) = x_0$. Define a mapping

$$\Phi : Deck(\tilde{X}/X) \to \pi_1(X, x_0)$$

as follows: Suppose $\sigma \in Deck(\tilde{X}/X)$ and v is a curve in X with initial point y_0 and end point $\sigma(y_0)$. (The homotopy class of v is uniquely determined since X is simply connected.) The curve $p \circ v$ in X has initial and end point x_0. Let $\Phi(\sigma)$ be the homotopy class of $p \circ v$.

1. Φ is a group homomorphism.

   Suppose $\sigma, \tau \in Deck(\tilde{X}/X)$ and v (resp. w) is a curve in X with initial point y_0 and end point $\sigma(y_0)$ (resp. $\tau(y_0)$). Then $\sigma \circ w$ is a curve with initial point $\sigma(y_0)$ and end point $\sigma\tau(y_0)$. Also $p \circ (\sigma \circ w) = p \circ w$. The product curve $v \cdot (\sigma \circ w)$ has initial point y_0 and end point $\sigma\tau(y_0)$. Thus

   $$\Phi(\sigma\tau) = [p \circ (v \cdot (\sigma \circ w))] = [p \circ v] \cdot [p \circ (\sigma \circ w)] = [p \circ v] \cdot [p \circ w] = \Phi(\sigma)\Phi(\tau).$$

2. Φ is bijective, this is left as an easy exercise.

□

Definition 2.1.24. Suppose X and Y are topological spaces, $p : Y \to X$ is a covering map and G is a subgroup of $Deck(Y/X)$. Two points $y, y' \in Y$ are called **equivalent modulo** G, if there exists $\sigma \in G$ such that $\sigma(y) = y'$. Clearly this really is an equivalence relation on Y.

Theorem 2.1.25 (Covering Spaces and Group Actions)**.** *Suppose X and Y are connected manifolds, $q : Y \to X$ is a covering map and $p : \tilde{X} \to X$ is the universal covering. Let $f : \tilde{X} \to Y$ be a continuous fiber-preserving mapping, which by the definition of the universal covering exists. Then f is a covering map and there exists a subgroup $G \subset Deck(\tilde{X}/X)$ such that two points $x, x' \in \tilde{X}$ are mapped onto the same point by f precisely if they are equivalent modulo G. Moreover $G \sim \pi_1(Y)$.*

Proof. For a detailed proof, see [GF12]. We just mention that once we have proved that f is a covering map, we can let $G = Deck(\tilde{X}/Y) \subset Deck(\tilde{X}/X)$ and use theorem 2.1.23 for the rest of the theorem. □

2.2 Group action and more examples of Riemann surfaces

This is a short section discussing how group actions give us examples of Riemann surfaces, we will see later as a corollary of theorem 2.5.1 that these examples are general in some sense. Suppose a group Γ acts on a Riemann surface X by holomorphic automorphisms. Suppose that the following two conditions hold:

- Around each point p of X, we can find an open neighbourhood N such that if $q_1, q_2 \in N$ and $\gamma \in \Gamma$ with $\gamma(q_1) = q_2$, then we must have $\gamma = 1$ and $q_1 = q_2$.
- Suppose p, q are points in X which do not lie in the same orbit of Γ. Then there are neighbourhoods N_1, N_2 of p, q, respectively, such that ΓN_1 is disjoint from N_2.

Then it is easy (like what we did in the construction of complex tori) to show that X/Γ is a Riemann surface with the quotient topology. (The two conditions above correspond to the complex charts and the Hausdorff property of X/Γ.)

Notice that the first condition above implies that Γ acts *freely* on X.

The important case of this quotient construction is when X is the upper halfplane $\mathbb{H}$. Of course, it is the same to consider the unit disc D. The holomorphic automorphisms of $\mathbb{H}$ are given by Möbius maps

$$z \to \frac{az + b}{cz + d},$$

where a, b, c, d are real and $ad - bc = 1$. In other words

$$Aut(\mathbb{H}) = PSL(2, \mathbb{R}).$$

If we have a suitable subgroup $\Gamma \subset PSL(2, \mathbb{H})$, we can construct a Riemann surface $\mathbb{H}/\Gamma$. Taking this over to $\mathbb{D}$, the holomorphic automorphisms are Möbius maps of the form

$$z \to \frac{\alpha z + \beta}{\bar{\beta} z + \bar{\alpha}},$$

where $|\alpha|^2 - |\beta|^2 = 1$. Now we have the following proposition:

Proposition 2.2.1. *Suppose that* Γ *is a discrete subgroup of* $Aut(\mathbb{H})$ *or* $Aut(\mathbb{D})$ *which acts* freely *on* $\mathbb{H}$ *or* $\mathbb{D}$*, respectively. Then the conditions above are satisfied.*

Proof. To prove this proposition, we work with the disc $\mathbb{D}$, which of course implies the other case.

To begin with, for the generalisation below, we will not use the assumption that Γ acts freely. The stabilizer of 0 in $Aut(\mathbb{D})$ is the circle, acting in the obvious way. This is just the case $\beta = 0$ in the general description above. If Γ is discrete, its intersection with this circle is a discrete subgroup of the circle, which must be a finite cyclic group. So the stabiliser of 0 in Γ is a finite cyclic group Z, say. It follows easily from the discreteness of Γ that we can choose a $k < 1$ such that if $\gamma \in \Gamma$ is $\gamma(z) = \dfrac{\alpha z + \beta}{\bar{\beta} z + \bar{\alpha}}$, where $|\alpha|^2 - |\beta|^2 = 1$, and if $|\beta| \leq k|\alpha|$, then γ is in Z. Now suppose η is a point in the disc with $|\eta| \leq \epsilon$ and that $|\gamma(\eta)| \leq c$. Here ϵ and c are poisitive real numbers to be determined. Then

$$|\alpha\eta + \beta| \leq c\left|\bar{\beta} z + \bar{\alpha}\right|,$$

Applying the triangle inequality twice, we get

$$|\beta| \leq c(|\alpha| + \epsilon|\beta|) + \epsilon|\alpha|.$$

Hence

$$|\beta| \leq \frac{\epsilon + c}{1 - c\epsilon}|\alpha|.$$

Now choose ϵ so small that

$$\frac{2\epsilon}{1 - c\epsilon} < k,$$

and let N be the disc of radius ϵ about 0. It follows that if $w_1, w_2 \in N$ and $\gamma \in \Gamma$ with $\gamma(w_1) = w_2$, then γ must lie in Z. So if Γ acts freely, as we now assume, γ must be the identity and the first condition is satisfied. This is for the particular point $0 \in \mathbb{D}$, but since $Aut(\mathbb{D})$ acts transitively on $\mathbb{D}$, the argument applies in the general case.

Now we turn to the second condition. Again, there is no loss in taking one of the points to be 0. Let q be any point in the disc not in the same Γ-orbit as 0. We take

$\tilde{N}_1, \tilde{N}_2$ to be the ϵ discs around 0 and q, respectively. If $\gamma \in \Gamma$ maps a point η in $\tilde{N}_1$ to $\tilde{N}_2$, then $|\gamma\eta| \leq c$ with $c = |q| + \epsilon$ and so, by the above, $|\beta| \leq k'|\alpha|$ with

$$k' = \frac{\epsilon + c}{1 - c\epsilon}.$$

We choose ϵ small enough that $k' < 1$.

The set of points $(\alpha, \beta) \in \mathbb{C}^2$ with $|\alpha|^2 - |\beta|^2 = 1$ and $|\beta| \leq k'|\alpha|$ is compact.

Since Γ is discrete, it follows that with these fixed $\tilde{N}_1, \tilde{N}_2$ there are only finitely many $\gamma \in \Gamma$ such that $\gamma(\tilde{N}_1) \cap \tilde{N}_2$ is non-empty. Now it is clear that we can find smaller neighbourhoods $N_i \subset \tilde{N}_i$ having the desired property. □

Having this proposition, we now investigate when the action of Γ is free. Suppose that a Möbius map in $PSL(2, \mathbb{R})$, as above, has a fixed point in the upper half-plane. This is the same as saying that there is a solution z of the equation

$$z = \frac{az + b}{cz + d},$$

with $Im(z) > 0$. If $c \neq 0$, the equation is equivalent to the quadratic equation

$$cz^2 + (d - a)z - b = 0,$$

so the condition is the same as saying that the discriminant is negative. Using the hypothesis $ad - bc = 1$, this discriminant is

$$(d - a)^2 + 4bc = (a + d)^2 - 4,$$

so we need $|a + d| < 2$. It is obvious that there is no fixed point in the case when $c = 0$ unless of course the map is the identity.

Now we introduce some famous examples from number theory. Fix a prime p and let $\tilde{\Gamma}_p$ be the set of 2×2 matrices M with integer entries, with determinant 1 and with $M = \pm id$ modulo p. Dividing by $\pm id$, we get a subgroup $\Gamma_p \subset PSL(2, \mathbb{R})$. For simplicity, we discuss here the case $p \geq 5$. Then it is clear that if a and d are equal to 1 modulo p, then the sum $a + d$ must have modulus more than 2. Thus Γ acts freely and we obtain a Riemann surface $X_p = \mathbb{H}/\Gamma_p$.

We will finish this section with a refinement of the quotient construction above. For *any* discrete subgroup $\Gamma \subset Aut(\mathbb{D})$ (or $Aut(\mathbb{H})$), the quotient has a natural Riemann surface structure. That is, the condition that Γ acts freely can be dropped. This is more surprising, but little extra work is required for the proof. What our analysis above gives is that if p is a point in D, then the stabilizer $Stab_p$ of p in Γ is a finite cyclic group, and there is a $Stab_p$-invariant neighborhood N of p which can be identified with a disc in such a way that the action of $Stab_p$ is the standard action of the cyclic subgroup of S^1. Then the quotient space $\mathbb{D}/\Gamma$ is covered by open sets which are either copies of N (for the points with a trivial stabilizer) or copies of proper quotients of N. But the map $z \mapsto z^r$

induces an identification between the quotient of the disc of radius ρ by the action of the cyclic group of order r and the disc of radius ρ^r . In this way we identify all members of this cover with discs, and these form an atlas of charts, making $\mathbb{D}/\Gamma$ into a Riemann surface.

More discussions about group actions will be made later.

2.3 Monodromy

Let $F : U \to V$ be a connected covering space of finite degree d, so that all points have exactly d preimages. If F corresponds to a subgroup $H \subset \pi_1(V, q)$, then the degree d is the index of the subgroup H.

Consider the fiber $F^{-1}(q)$ over q. Denote the d points in this fiber by $\{x_1, ..., x_d\}$. Every loop γ in V based at q can be lifted to d paths $\tilde{\gamma}_1, \ldots, \tilde{\gamma}_d$, where $\tilde{\gamma}_i$ is the unique lift of γ which starts at x_i. In other words, $\tilde{\gamma}_i(0) = x_i$, for every i.

Next consider the endpoints $\tilde{\gamma}_i(1)$; these also lie over q, and indeed form the entire preimage set $F^{-l}(q)$. Hence each is an x_j for some j; we denote $\tilde{\gamma}_i(1)$ by $\sigma(i)$.

This function σ is a permutation of the indices $\{l, ..., d\}$, and it is easy to see that it depends only on the homotopy class of the loop γ. Therefore we have a group homomorphism

$$\rho : \pi_1(V, q) \to S_d,$$

where S_d denotes the symmetric group of all permutations on d indices.

Definition 2.3.1. The **monodromy representation** of a covering map $F : U \to V$ of finite degree d is the group homomorphism $\rho : \pi_1(V, q) \to S_d$ defined above.

By connectivity of U we know that the image $H \subset S_d$ of ρ is *transitive*.

For a (non-constant) holomorphic map $F : X \to Y$, as we did in section 2.1, the map $F : X' \to Y'$ is a covering map, and thus has a monodromy representation, which is called the **monodromy representation** of the holomorphic map. Since X is connected so is X', and thus the image of the monodromy map is a transitive subgroup of S_d.

For each branch point $b \in Y$, choose a small open neighborhood W of b in Y; the punctured open set $W - \{b\}$ is an open subset of V, isomorphic to a small punctured disc. Denote by $u_1, \ldots, u_k$ the k preimages of b in X; the number k will be no greater than the degree d of F since b is a branch point, so that at least one of the u_j's are ramifcation points. We choose W small enough so that $F^{-1}(W)$ decomposes as a disjoint union of open neighborhoods $U_1, \ldots, U_k$ of the points $u_1, \ldots, u_k$ respectively. Set $m_j = \text{mult}_{u_j}(F)$ to be the multiplicity of F at these preimage points; by 1.1.9, there are coordinates z_j on the U_j's and z on W so that the map F has the form $z = z_j^{m_j}$ on U_j.

Now consider $U_j - \{u_j\}$. $U_j - \{u_j\}$ is isomorphic to a punctured disc, and the map F sends $U_j - \{u_j\}$ to $W - \{b\}$ via the mth power map. Choose a path α from the base point q to a point q_0 in $W - \{b\}$, and a loop β in $W - \{b\}$ based at q_0 with winding

number one around the branch point b. Then the path $\alpha^{-1}\beta\alpha$ (composing paths from right to left) is a loop γ on V based on q. We call such a loop on V a **small loop** on V around b.

In analyzing how the small loop γ lifts to the covering $F : U \to V$, it is clear that traversing the path α simply gives an identification of the fiber of F over q with the fiber of F over q_0; following the reverse path α^{-1} gives the inverse identification. Therefore the permutation σ of the fiber of F over q which is induced (via the monodromy representation ρ) by the small loop γ around b is actually determined up to this identification by the loop β around the branch point b.

Above the open set $W - \{b\}$ we have k punctured discs $U_j - \{u_j\}$, each mapping to $W - \{b\}$ via a power map. The monodromy for each cover $U_j - \{u_j\} \to W - \{b\}$ is a cyclic permutation of those m_j preimages of q_0 which lie in U_j. In fact the loop β induces a cyclic permutation of these points, and therefore the loop γ also maps to a cyclic permutation of the corresponding identified points in the fiber above the base point g. Therefore we know the cycle structure of the permutation σ, and we have proved the following:

Lemma 2.3.2. *Suppose that above the branch point $b \in Y$ there are k preimages $u_1, \ldots, u_k$, with $\mathrm{mult}_{u_j}(F) = m_j$. Then with the above notation the cycle structure of the permutation σ representing a small loop around b (after the identifcation via the path α) is $(m_1, \ldots, m_k)$.*

Suppose a connected real manifold V is given, with a chosen base point q. Suppose further that we have a group homomorphism $\rho : \pi_1(V, q) \to S_d$, from the fundamental group of V to a symmetric group S_d, with a *transitive* image. Fix an index, say 1. Let $H \subset \pi_1(V, q)$ be the subgroup consisting of those homotopy classes $[\gamma]$ such that $\rho([\gamma])$ fixes the index 1:

$$H := \{[\gamma] \in \pi_1(V, q) : \rho([\gamma])(1) = 1\}$$

Then H has index d in $\pi_1(V, q)$ (by the orbit-stabilizer theorem), and by the discussion in the last section (please check that the action of H on the universal covering satisfies the two conditions mentioned in the last section) induces a connected covering space $F_\rho : U_\rho \to V$. Moreover this covering has the property that its monodromy representation is exactly the given homomorphism ρ by the definition of U_ρ as quotient of the space of paths.

This process essentially gives an inverse to the mapping which sends a covering to its monodromy representation; the only caveat is that this only works for coverings of *finite* degree. Hence we have the following: for a connected real manifold V,there is a 1-1

correspondence

$$\left\{\begin{array}{c}\text{isomrophism classes of}\\ \text{connected coverings}\\ F: U \to V\\ \text{of degree } d\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\text{group homomorphisms}\\ \rho: \pi_1(V,q) \to S_d\\ \text{with transitive image}\\ \text{up to conjugacy in } S_d\end{array}\right\}$$

Using theorem theorem 2.1.4 we can obtain a version of *umbranched* holomorphic covering maps, while what we are really interested is the version of *branched* holomorphic covering maps discussed below.

Let Y be a compact Riemann surface, and let $B \subset Y$ be a finite subset. Let $V = Y - B$ be the complement of B, which is an open subset of Y and is also a Riemann surface. Fix a base point $q \in V$. Suppose that one has a group homomorphism $\rho: \pi_1(V,q) \to S_d$ with transitive image. Let $F_\rho: U_\rho \to V$ be the covering map induced by ρ; the space U_ρ is a Riemann surface and the map F_ρ is a holomorphic map of degree d, by the previous discussion.

Now we prove two theorems about extending umbranched coverings:

Theorem 2.3.3 (Existence of Extension)**.** *Suppose X is a Riemann surface, $A \subset X$ is a closed discrete subset and let $X' = X - A$. Suppose Y' is another Riemann surface and $\pi': Y' \to X'$ is a proper* unbranched *holomorphic covering. Then π' extends to a* branched *covering of X, i.e., there exists a Riemann surface Y, a proper holomorphic mapping $\pi: Y \to X$ and a fiber-preserving biholomorphic mapping*

$$\varphi: (Y - \pi^{-1}(A)) \to Y'.$$

Theorem 2.3.4 (Uniqueness of Extension)**.** *Suppose X, Y and Z are Riemann surfaces and $\pi: Y \to X, \tau: Z \to X$ are proper holomorphic covering maps. Let $A \subset X$ be a closed discrete subset and let $X' := X - A, Y' := \pi^{-1}(X')$ and $Z' := \tau^{-1}(X')$. Then every fiber-preserving biholomorphic mapping $\phi': Y' \to Z'$ can be extended to a fiber-preserving biholomorphic mapping $\phi: Y \to Z$. In particular every covering transformation $\sigma' \in Deck(Y'/X')$ can be extended to a covering transformation$\phi \in Deck(Y/X)$.*

Remark. Please figure out how the name of this theorem comes.

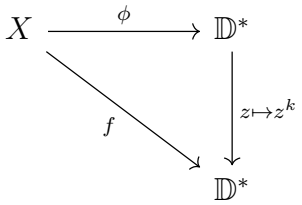
Proof.

- Existence:

  We first note that the theorems hold when $X = \mathbb{D}$ and $A = \{0\}$. Here we just list this as a lemma:

  Lemma 2.3.5. *Suppose X is a Riemann surface, $\mathbb{D}$ is the unit disk and $f: X \to \mathbb{D}^*$ is a* proper *non-constant holomorphic map which is unbranched over $\mathbb{D}^* = \mathbb{D} - \{0\}$. Then there exists a natural number $k \geq 1$ and a biholomorphic mapping $\phi: X \to \mathbb{D}$*

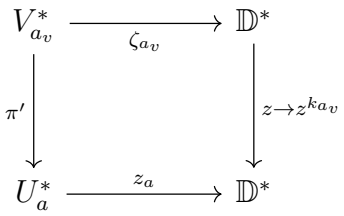
such that the diagram



commutes.

Proof. For a proof of this one can just use theorem 2.1.25 first to classify all the covering spaces of $\mathbb{D}^*$ (which turns out to be $z^k : \mathbb{D}^* \to \mathbb{D}$ or $exp : \mathbb{H} \to \mathbb{D}$) and then see the result directly. See [GF12] for details. □

Going back to our proof, for every $a \in A$ choose a coordinate neighborhood (U_a, z_a) on X with the following properties: $z_a(a) = 0, z_a(U_a)$ is the unit disk in $\mathbb{C}$ and $U_a \cap U_{a'} = 0$ if $a \neq a'$. Let $U_a^* := U_a - \{a\}$. Since $\pi' : Y' \to X'$ is proper, $\pi'^{-1}(U_a^*)$ consists of a finite number of connected components $V_{a_v}^*, v = 1, ..., n(a)$. For every v the mapping $\pi' : V_{a_v}^* \to U_a^*$ is an unbranched covering. Let its covering number be k_{a_v}. By our lemma there exist biholomorphic mappings $\zeta_{a_v} : V_{a_v}^* \to \mathbb{D}^*$ of $V_{a_v}^*$ onto the punctured unit disk $\mathbb{D}^* = \mathbb{D} - \{0\}$ such that the diagram



commutes.

Now choose "ideal points" $p_{a_v}, a \in A, v = 1, ..., n(a)$, i.e., pairwise distinct elements of some set disjoint from Y'. Then on

$$Y := Y' \sqcup \{p_{a_v} : a \in A, v = 1, \ldots, n(a)\}$$

there exists precisely one topology with the following property:

If $W_i, i \in I$ is a neighborhood basis of a, then

$$\{p_{a_v}\} \sqcup (\pi'^{-1}(W_i) \cap V_{a_v}^*)$$

is a neighborhood basis of p_{a_v} and on Y' it induces the given topology. This makes Y into a Hausdorff space. Define $\pi : Y \to X$ by $\pi(y) = \pi'(y)$ for $y \in Y'$ and $\pi(p_{a_v}) = a$. Then, as one easily checks, π is proper.

In order to make Y into a Riemann surface, add to the charts of the complex structure of Y' the following charts. Let $V_{a_v} := V_{a_v}^* \sqcup \{p_{a_v}\}$ and let

$$\zeta_{a_v} : V_{a_v} \to \mathbb{D}$$

be the continuation of the mapping $\zeta_{a_v} : V_{a_v}^* \to \mathbb{D}^*$ described above which is obtained by defining $\zeta_{a_v}(p_{a_v}) := O$. Since the last mapping is biholomorphic with respect to the complex structure of Y', the new charts $\zeta_{a_v} : V_{a_v} \to \mathbb{D}$ are holomorphically compatible with the charts of the complex structure of Y'.

The mapping $\pi : Y \to X$ is holomorphic. Since $(Y - \pi^{-1}(A)) = Y'$ by construction, we may choose $\phi : (Y - \pi^{-1}(A)) \to Y'$ to be the identity mapping. This then shows the existence of a continuation of the covering $\pi' : Y' \to X'$.

- Uniqueness:

  Suppose $a \in A$ and (U, z) is a coordinate neighborhood of a such that $z(a) = 0$ and $z(U)$ is the unit disk. Let $U^* = U - \{a\}$. Moreover we may assume that U is so small that π and τ are unbranched over U^*. Let $V_1, \ldots, V_n$ (resp. $W_1, \ldots, W_m$) be the connected components of $\pi^{-1}(U)$ (resp. $\tau^{-1}(U)$). Then $V_(^*v) := (V_v - \pi^{-1}(a))$ (resp. $W_u^* = (W_u - \tau^{-1}(a))$) are the connected components of $\pi^{-1}(U^*)$ (resp. $\tau^{-1}(U*)$).

  Since $\sigma' : \pi^{-1}(U^*) \to \tau^{-1}(U^*)$ is biholomorphic, $n = m$ and one may renumber so that $\sigma'(V_v^*) = W_v^*$. Since $\pi : V_v^* \to U^*$ is a finite sheeted unbranched covering, $V_v \cap \pi^{-1}(a)$ (resp. $W_v \cap \tau^{-1}(a)$) consists by our lemma above of exactly one point b_v (resp. c_v). Hence $\sigma' : \pi^{-1}(U^*) \to \tau^{-1}(U^*)$ can be continued to a bijective mapping $\pi^{-1}(U) \to \tau^{-1}(U)$ which assigns to b_v the point c_v. Since $\pi : V_v \to U$ and $\tau : W_w \to U$ are proper (thus closed), the continuation is a homeomorphism and by Riemann's Removable Singularities Theorem (theorem 1.1.5) it is biholomorphic as well. (The Removable Singularities Theorem applies since V_v and W_v are isomorphic to the unit disk.) If one now applies this construction to every exceptional point $a \in A$, then one gets the desired continuation $\sigma : Y \to Z$.

□

With these theorems in mind we continue our description of *branched* coverings via monodromy maps. We can extend our map $F_\rho : U_\rho \to V = Y - B$ uniquely. Since $F_\rho : U_\rho \to V$ is in one-one correspondence with group homomorphisms $\rho : \pi_1(Y - B, q) \to S_d$ with transitive image up to conjugation, we obtain the following one-one correspondence:

$$\left\{\begin{array}{c}\text{isomrophism classes of}\\ \text{holomorphic maps}\\ F : X \to Y\\ \text{of degree } d\\ \text{whose branch point lie in } B\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\text{group homomorphisms}\\ \rho : \pi_1(Y - B, q) \to S_d\\ \text{with transitive image}\\ \text{up to conjugacy in } S_d\end{array}\right\}$$

Moreover at a point $b \in B$, if γ is a small loop in $Y - B$ around b based at q, and if $\rho([\gamma])$ has cycle structure $(m_1, \ldots, m_k)$, then there are k preimages $u_1, \ldots, u_k$ of b in the corresponding cover $F_\rho : X_\rho \to Y$, with $\text{mult}_{u_j}(F_\rho) = m_j$ for each j.

Example (Hyperelliptic Curves)**.** *Let X be a compact Riemann surface. Suppose that $F : X \to \mathbb{CP}^1$ is a holomorphic map of degree 2. Then X is a hyperelliptic curve.*

Proof. Let g be the genus of X. By Hurwitz's formula (See theorem 3.2.17 or [Mir95]) we have that the number of branch points of F is $2g+2$, and since the degree of F is 2, each branch point has as its preimage a single ramification point with multiplicity two.

Let $B = b_1, \ldots, b_{2g+2}$ be the set of branch points of F, and let $V = \mathbb{CP}^1 - B$. The monodromy representation for F is a group homomorphism $\rho : \pi_1(V) \to S_2$ and we denote by σ_i the image under ρ of the homotopy class of a small loop in V around b_i.

Since there is one point of multiplicity two lying above b_i for each i, the cycle structure of σ_i, must be (2) for each i. This forces σ_i to be the transposition $(12) \in S_2$ for each i. Hence once the branch points $b_1, \ldots, b_{2g+2}$ are chosen, there is no choice for the permutations σ_i, and the Riemann surface X is determined up to isomorphism by the branch points alone.

On the other hand, if z is an affine coordinate on $\mathbb{CP}^1$, and none of the branch points b_i is the point at infinity, then the hyperelliptic curve Y defined by the equation $y^2 = \prod_{i-1}^{2g+2}(z - b_i)$ also covers $\mathbb{CP}^1$ with the same branch points by a map of degree 2 (given by the function z). Therefore this covering $z : Y \to \mathbb{CP}^1$ has the same branch points and the same monodromy as does the given map $F : X \to \mathbb{CP}^1$. By the 1-1 correspondence, we must have that these coverings are isomorphic,and so $X \cong Y$.

If one of the branch points, say b_{2g+2}, is the point at infinity, simply consider instead the hyperelliptic curve defined by the equation $y^2 = \prod_{i=1}^{2g+1}(z - b_i)$. The argument proceeds in the same manner.

Also by this monodromy argument we can see the fact expressed in ?? about the construction of hyperelliptic surfaces is true. Let's see this figure again here. □

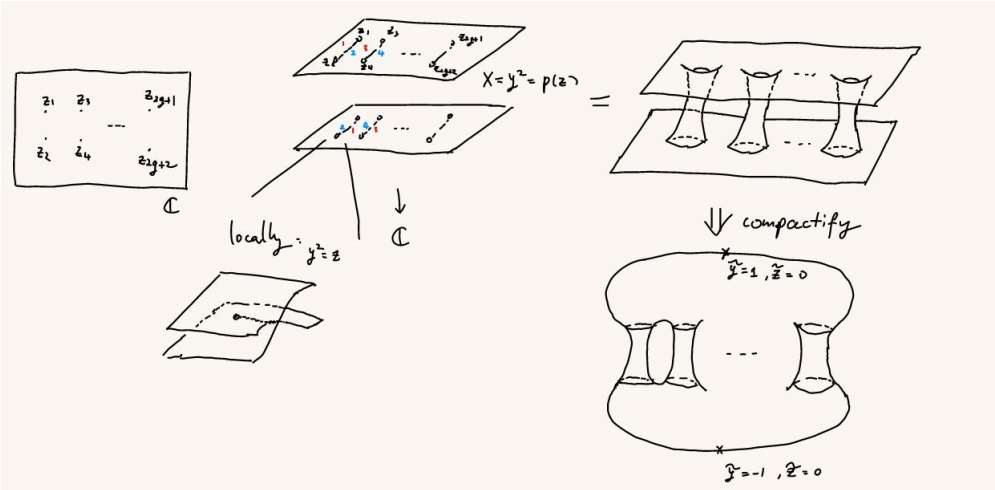


Figure 2.1: Visual construction of hyperelliptic curves

2.4 Galois theory: algebraic functions

Suppose X and Y are Riemann surfaces, $\pi : Y \to X$ is an n-sheeted unbranched holomorphic covering map and f is a meromorphic function on Y. Every point $x \in X$ has an open neighborhood U such that $\pi^{-1}(U)$ is the disjoint union of open sets $V_1, \ldots, V_n$ and $\pi : V_v \to U$ is biholomorphic for $v = 1, \ldots, n$. Let $\tau_v : U \to V_v$ be the inverse mapping of $\pi | V_v \to U$ and let $f_v := \tau_v^* f = f \circ \tau_v$. Suppose T is an indeterminate and consider

$$\prod_{v=1}^{n} (T - f_v) = T^n + c_1 T^{n-1} + \cdots + c_n.$$

Then the c_v are meromorphic functions in U and

$$c_v = (-1)^v s_v(f_1, \ldots, f_n),$$

where s_v denotes the v-th elementary symmetric function in n variables. If one carries out this same construction in a neighborhood U' of another point $x' \in X$, then one gets the same functions $c_1, \ldots, c_n$. Thus these functions piece together to give global meromorphic functions $c_1, \ldots, c_n \in \mathscr{M}(X)$, which we call the elementary symmetric functions of f with respect to the covering $Y \to X$.

This construction works for branched holomorphic maps as well.

Theorem 2.4.1. *Suppose X and Y are Riemann surfaces and $\pi : Y \to X$ is an n-sheeted branched holomorphic covering map. Suppose $A \subset X$ is a closed discrete subset which contains all the critical values of π and let $B = \pi^{-1}(A)$. Suppose f is a holomorphic (resp. meromorphic) function on $Y \backslash B$ and $c_1, \ldots, c_n \in \mathscr{O}(X \backslash A)$ (resp. $\in \mathscr{M}(X \backslash A)$) are the elementary symmetric functions of f. Then f may be continued holomorphically (resp. meromorphically) to Y precisely if all the c_v may be continued holomorphically (resp. meromorphically) to X.*

The Theorem ensures that the elementary symmetric functions of a function $f \in \mathscr{M}(Y)$ are also defined when the map $Y \to X$ is a branched holomorphic covering.

Proof. Suppose $a \in A$ and $b_1, \ldots, b_m$ are the preimages of a. Suppose (U, z) is a relatively compact (i.e has compact closure) coordinate neighborhood of a with $z(a) = 0$ and $U \cap A = \{a\}$. Then $V := \pi^{-1}(U)$ is a relatively compact neighborhood of each of the b_μ.

1. First consider the case $f \in \mathscr{O}(Y \setminus B)$.

(a) Assume f may be continued holomorphically to all the points b_μ. Then f is bounded on $V \setminus \{b_1, \ldots, b_m\}$. This implies that all the c_v are bounded on $U \setminus \{a\}$. By Riemann's Removable Singularities Theorem, they may all be continued holomorphically to a.

(b) Suppose all the c_v may be continued holomorphically to a. Then all the c_v are bounded on $U \setminus \{a\}$. But this implies f is bounded on $V \setminus \{b_1, \ldots, b_m\}$, for, if $y \in V \setminus \{b_1, \ldots, b_m\}$ and $x = \pi(y)$, then

$$f(y)^n + c_1(x)f(y)^{n-1} + \cdots + c_n(x) = 0.$$

Again, Riemann's Removable Singularities Theorem implies that f may be continued holomorphically to every point b_μ.

2. Now suppose $f \in \mathscr{M}(Y \setminus B)$.

(a) Assume f may be continued meromorphically to all points b_μ. The function $\varphi := \pi^* z \in \mathscr{O}(V)$ vanishes at all the points b_μ. Thus $\varphi^k f$ may be continued holomorphically to all the points b_μ if k is sufficiently large. The elementary symmetric functions of $\varphi^k f$ are $z^{kv}c_v$, and by the first part of the proof, they may be continued holomorphically to a. Thus all the c_v may be continued meromorphically to a.

(b) Suppose all the c_v may be continued meromorphically to a. Using the above notation, one has: For k sufficiently large, all the $z^{kv}c_v$ admit holomorphic continuations to a. Thus $\varphi^k f$ admits a holomorphic continuation to all the points b_μ. This implies that f may be continued meromorphically to all of the points b_μ. □

Remark. Through this proof we have used Riemann's removable singularities theorem repeatedly, this is in fact a reflection of the fact that a Riemann surface is *smooth*. For example, if we consider the algebraic curve $C := \{y^2 = x^3\} \subset \mathbb{C}^2$, then the rational function

$$f := \frac{y}{x}$$

is locally bounded, but it cannot be extended to a polynomial function on C as the curve is not isomorphic to the affine line.

We can somehow understand the removable singularity theorem as saying that Riemann surfaces are **normal** in the sense that every meromorphic (rational) function that is *integral* over the ring of holomorphic (polynomial) functions is actually holomorphic (polynomial).

For example, the meromorphic function $\dfrac{y}{x}$ is actually integral as $f^2 - x = 0$ on C, so the curve is not *normal*. Geometrically we can see that this curve has a "wedge point" at which the curve is not smooth.

For more algebro-geometric discussion about this, see [Loo20]

For later use, note that the proof does not use the fact that Y is connected. Thus the Theorem also holds in the case that Y is a disjoint union of finitely many Riemann surfaces.

If $\pi\colon Y \to X$ is a non-constant holomorphic map between Riemann surfaces X and Y, then for any meromorphic function f on X, the function $\pi^* f := f \circ \pi$ is a meromorphic function on Y. Thus there is a map

$$\pi^*\colon \mathscr{M}(X) \to \mathscr{M}(Y)$$

which is a monomorphism of fields.

Theorem 2.4.2. *Suppose X and Y are Riemann surfaces and $\pi : Y \to X$ is a branched holomorphic n-sheeted covering map. If $f \in \mathscr{M}(Y)$ and $c_1, \ldots, c_n \in \mathscr{M}(X)$ are the elementary symmetric functions of f, then*

$$f^n + (\pi^* c_1) f^{n-1} + \cdots + (\pi^* c_{n-1}) f + \pi^* c_n = 0.$$

The monomorphism $\pi^* : \mathscr{M}(X) \to \mathscr{M}(Y)$ is an algebraic field extension of degree $\leq n$. Moreover, if there exist an $f \in \mathscr{M}(Y)$ and an $x \in X$ with preimages $y_1, \ldots, y_n \in Y$ such that the values $f(y_v)$ for $v = 1, \ldots, n$ are all distinct, then the field extension $\pi^* : \mathscr{M}(X) \to \mathscr{M}(Y)$ has degree n.

Proof. The existence of the equation

$$f^n + \sum_{v=1}^{n} (\pi^* c_v) f^{n-v} = 0$$

follows directly from the definition of the elementary symmetric functions of f.

Let $L := \mathscr{M}(Y)$ and $K := \pi^* \mathscr{M}(X) \subset L$. Then every $f \in L$ is algebraic over K, and the minimal polynomial of f over K has degree $\leq n$. Suppose $f_0 \in L$ is an element for which the degree n_0 of its minimal polynomial is maximal. We claim $L = K(f_0)$. Choose an arbitrary element $f \in L$ and consider the field $K(f_0, f)$. By the Theorem of the Primitive Element, there exists $g \in L$ such that $K(f_0, f) = K(g)$. (The finite separable extension is contained in its normal closure, which is Galois, and therefore it has only finitely many intermediate fields by Galois correspondence.) By the definition of n_0, one has $\dim_K K(g) \leq n_0$. On the other hand,

$$\dim_K K(f_0, f) \geq \dim_K K(f_0) = n_0.$$

Thus $K(f_0) = K(f_0, f)$, and $f \in K(f_0)$.

Finally, if the degree of the minimal polynomial of f over K were equal to $m < n$, then f would be able to take at most m different values over every point $x \in X$. □

Definition 2.4.3. Suppose X and Y are Riemann surfaces and $\pi : Y \to X$ is a branched holomorphic covering. Let $A \subset X$ be the set of critical values of π and let $X' := X \setminus A$ and $Y' := \pi^{-1}(X')$. Then the covering $Y \to X$ is called **Galois** if the covering $Y' \to X'$ is Galois.

As a corollary of theorem 1.1.2, we have:

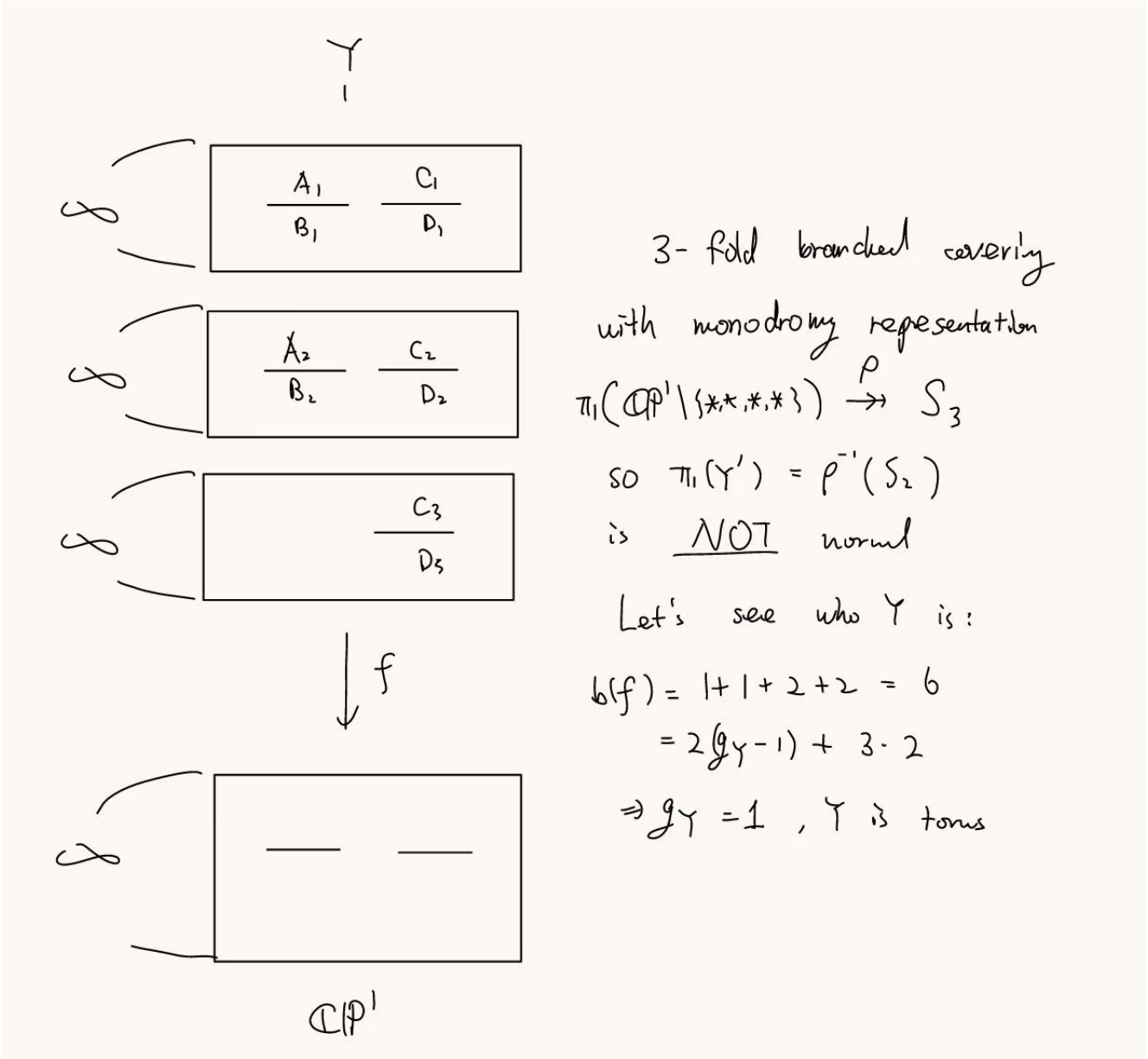


Figure 2.2: A non-Galois branched covering

Proposition 2.4.4. *Let $\mathscr{O}_x$ be the ring of holomorphic function germs at a point x of a Riemann surface and let*

$$P(T) = T^n + c_1 T^{n-1} + \cdots + c_n \in \mathscr{O}_x[T].$$

Suppose that the polynomial

$$p(T) := T^n + c_1(x)T^{n-1} + \cdots + c_n(x) \in \mathbb{C}[T]$$

has n distinct zeros $w_1, \ldots, w_n$. Then there exist elements $\varphi_1, \ldots, \varphi_n \in \mathscr{O}_x$ such that $\varphi_v(x) = w_v$ and

$$P(T) = \prod_{v=1}^{n} (T - \varphi_v).$$

Theorem 2.4.5. *Suppose X is a Riemann surface and*

$$P(T) = T^n + c_1 T^{n-1} + \cdots + c_n \in \mathscr{M}(X)[T]$$

is an irreducible polynomial of degree n. Then there exist a Riemann surface Y, a branched holomorphic n-sheeted covering $\pi\colon Y \to X$, and a meromorphic function $F \in \mathscr{M}(Y)$ such that $(\pi^* P)(F) = 0$. The triple (Y, π, F) is uniquely determined in the following sense. If (Z, τ, G) has the corresponding properties, then there exists exactly one fiber-preserving biholomorphic mapping $\sigma : Z \to Y$ such that $G = \sigma^* F$.

To simplify the terminology, (Y, π, F) is called the **algebraic function** defined by the polynomial $P(T)$.

Remark. The classical case is when X is the Riemann sphere $\mathbb{CP}^1$. Then the coefficients c_v of the polynomial $P(T)$ are rational functions in one variable. Since $\mathbb{CP}^1$ is compact and $\pi : Y \to \mathbb{CP}^1$ is proper.

Proof. Let $\Delta \in \mathscr{M}(X)$ be the discriminant of the polynomial $P(T)$. (Δ is a certain polynomial in the coefficients of P.) The discriminant cannot vanish identically, for otherwise P would be reducible. There exists a closed discrete subset $A \subset X$ such that at every point $x \in X' := X \setminus A$, all the functions $c_1, \ldots, c_n$ are holomorphic and $\Delta(x) \neq 0$. Then for every $x \in X'$, the polynomial

$$p_x(T) := T^n + c_1(x)T^{n-1} + \cdots + c_n(x) \in \mathbb{C}[T]$$

has n distinct zeros. Now we will use the topological space $|\mathscr{O}| \to X$ associated to the sheaf $\mathscr{O}$. Let $Y' \subset |\mathscr{O}|$ be the set of all the function germs $\varphi \in \mathcal{O}_x$, $x \in X'$, which satisfy the equation $P(\varphi) = 0$, and let $\pi' : Y' \to X'$ be the canonical projection. By proposition 2.4.4, for every point $x \in X'$, there exist an open neighborhood $U \subset X'$ and holomorphic functions $f_1, \ldots, f_n \in \mathscr{O}(U)$ such that

$$P(T) = \prod_{v=1}^{n}(T - f_v) \quad \text{on } U.$$

Then $\pi'^{-1}(U) = \bigcup_{v=1}^{n}[U, f_v]$. The $[U, f_v]$ are disjoint, and $\pi'|_{[U,f_v]} : [U, f_v] \to U$ is a homeomorphism. This shows that $Y' \to X'$ is a covering map. The connected components of Y' are Riemann surfaces which also admit covering maps over X'. Let $f : Y' \to \mathbb{C}$ be defined by $f(\varphi) := \varphi(\pi'(\varphi))$. Then f is holomorphic, and by construction,

$$f(y)^n + c_1(\pi'(y))f(y)^{n-1} + \cdots + c_n(\pi'(y)) = 0$$

for every $y \in Y'$. By theorem 2.3.3, the covering $\pi' : Y' \to X'$ may be continued to a proper holomorphic covering $\pi : Y \to X$, where we identify Y' with $\pi^{-1}(X')$. By theorem 2.4.1, f may be extended to a meromorphic function $F \in \mathscr{M}(Y)$, for which

$$(\pi^* P)(F) = F^n + (\pi^* c_1)F^{n-1} + \cdots + \pi^* c_n = 0.$$

Now we will show that Y is connected and thus a Riemann surface. Suppose this is not the case. Then Y has finitely many connected components $Y_1, \ldots, Y_k$, and $\pi|_{Y_i} : Y_i \to X$ is a proper holomorphic n_i-sheeted covering, where

$$\sum n_i = n.$$

Using the elementary symmetric functions of $F|Y_i$, one gets polynomials $P_i(T) \in \mathscr{M}(X)[T]$ of degree n_i such that

$$P(T) = P_1(T)P_2(T)\cdots P_k(T).$$

But this contradicts the assumption that $P(T)$ is irreducible.

Uniqueness. Suppose (Z, τ, G) is another algebraic function defined by the polynomial $P(T)$. Let $B \subset Z$ be the union of the poles of G and the branch points of τ, and let $A' := \tau(B)$. Let

$$X'' := X' \setminus A', \quad Y'' := \pi^{-1}(X''), \quad Z'' := \tau^{-1}(X'').$$

Define a fiber-preserving mapping $\sigma'' : Z'' \to Y''$ in the following way. Let $z \in Z''$, $\tau(z) = x$, and $\varphi \in \mathscr{O}_x$ be the function germ $\varphi := \tau_* G_z$. Then $P(\varphi) = 0$. By the construction of Y', one sees that φ is a point of Y' over x and thus $\varphi \in Y''$. Set $\sigma''(z) = \varphi$.

From the definition, it follows directly that σ'' is continuous. Since σ'' is fiber-preserving, σ'' is thus holomorphic. Moreover, σ'' is proper since $\pi : Y'' \to X''$ is continuous and $\tau : Z'' \to X''$ is proper. Hence σ'' is surjective. Because $Y'' \to X''$ and $Z'' \to X''$ have the same number of sheets, $\sigma'' : Z'' \to Y''$ is biholomorphic. Also, from the definition of σ'', one gets $G|_{Z''} = (\sigma'')^*(F|_{Y''})$. By theorem 2.3.4, σ'' can be extended to a fiber-preserving biholomorphic mapping $\sigma : Z \to Y$ for which one then has $G = \sigma^* F$.

The mapping σ is in fact uniquely determined by the property $G = \sigma^* F$. For, otherwise, there would exist a covering transformation $\alpha : Y \to Y$ different from the identity such that $\alpha^* F = F$. But this is not possible since F assumes distinct values on the fiber $\pi^{-1}(x)$ over every point. □

Example. *For $f(z) = \prod_{i=1}^{n}(z - a_i)$ on $\mathbb{CP}^1$, where a_i are distinct, the polynomial*

$$T^2 - f(z) \in \mathscr{M}(\mathbb{CP}^1)[T]$$

is irreducible, and the algebraic function defined by it is just given by the hyperelliptic surface it produces.

If X and Y are Riemann surfaces and $\pi\colon Y \to X$ is a branched holomorphic covering map, then $\mathrm{Deck}(Y/X)$ has a representation into the automorphism group of the field $\mathscr{M}(Y)$ defined in the following way. For $\sigma \in \mathrm{Deck}(Y/X)$, let $\sigma f := f \circ \sigma^{-1}$. Clearly, the correspondence $f \mapsto \sigma f$ is an automorphism of $\mathscr{M}(Y)$. The mapping

$$\mathrm{Deck}(Y/X) \to \mathrm{Aut}(\mathscr{M}(Y))$$

is a group homomorphism. For suppose $\sigma, \tau \in \mathrm{Deck}(Y/X)$. Then for every $f \in \mathscr{M}(Y)$,

$$(\sigma\tau)f = f \circ (\sigma\tau)^{-1} = f \circ \tau^{-1} \circ \sigma^{-1} = \sigma(f \circ \tau^{-1}) = \sigma(\tau f).$$

Trivially, every such automorphism $f \mapsto \sigma f$ leaves invariant the functions of the subfield $\pi^*\mathscr{M}(X) \subset \mathscr{M}(Y)$ and thus is an element of the Galois group $\mathrm{Aut}(\mathscr{M}(Y)/\pi^*\mathscr{M}(X))$.

Theorem 2.4.6 ("Fundamental theorem of Galois theory")**.** *Suppose X is a Riemann surface, $K := \mathscr{M}(X)$ is the field of meromorphic functions on X, and $P(T) \in K[T]$ is an irreducible monic polynomial of degree n. Let (Y, π, F) be the algebraic function defined by $P(T)$, and let $L = \mathscr{M}(Y)$. By means of the monomorphism $\pi^* : K \to L$, consider K as a subfield of L. Then $L : K$ is a field extension of degree n, and $L \cong K[T]/(P(T))$. Every covering transformation $\sigma : Y \to Y$ of Y over X induces an automorphism $f \mapsto \sigma f := f \circ \sigma^{-1}$ of L leaving K fixed, and the mapping*

$$Deck(Y/X) \to Aut(L/K)$$

which is so defined, is a group isomorphism. The covering $Y \to X$ is Galois precisely if the field extension $L : K$ is Galois.

Proof. The fact that $L : K$ is a field extension of degree n follows from the last statement of theorem 2.4.2. Since $P(F) = 0$, there is a homomorphism $K[T]/(P(T)) \to L$. Since both these fields are of degree n over K, this is an isomorphism.

The mapping $\mathrm{Deck}(Y/X) \to \mathrm{Aut}(L/K)$ is injective, because $\sigma F \neq F$ for any covering transformation σ which is not the identity.(F has distinct values at points of the

fiber over all but a discrete set of points.) This mapping is also surjective. For, suppose $\alpha \in \mathrm{Aut}(L/K)$. Then $(Y, \pi, \alpha F)$ is also an algebraic function defined by the polynomial $P(T)$. Thus, by the uniqueness statement of theorem 2.4.5, there exists a covering transformation $\tau \in \mathrm{Deck}(Y/X)$ such that $\alpha F = \tau^* F$. If $\sigma := \tau^{-1}$, then

$$\sigma F = F \circ \sigma^{-1} = F \circ \tau = \tau^* F = \alpha F.$$

Since L is generated by F over K, the automorphism $f \mapsto \sigma f$ of L coincides with α.

The last statement of the Theorem follows from the fact that Y is Galois over X (resp. L is Galois over K) precisely when $\mathrm{Deck}(Y/X)$ (resp. $\mathrm{Aut}(L/K)$) contains n elements. (For covering spaces to be Galois, by uniqueness of lifting we know the stabilizer of any point is trivial, thus the orbit-stabilizer theorem gives the equivalence.) □

Thus the study of (finite-sheeted) holomorphic maps between Riemann surfaces corresponds to the Galois theory studying the (finite) algebraic extensions of function fields. To make this concrete, let's prove in the following that the function field actually determines a Riemann surface.

Definition 2.4.7. Let K be a field, K^* is the multiplicative group of it. If a group homomorphism

$$v : K^* \to \mathbb{Z}$$

is surjective, and

$$v(f+g) \geq \min\{v(f), v(g)\}, \forall f, g \in K^*,$$

then v is called a **discrete valuation** on K.

Usually we define $v(0) = \infty$. For any point $p \in X$ where X is a Riemann surface, the map $v_p(f) := v(p, f) = \mathrm{ord}_p(f)$ gives a discrete valuation. A discrete valuation satisfies the following properties:

(1) $v(1) = 0, v(-1) = 0, v(f) = v(-f)$.

(2) When $v(f) \neq v(g)$, $v(f+g) = \min\{v(f), v(g)\}$.

(3) If $\mathbb{C} \subset K$, then $v(c) = 0, \forall c \in \mathbb{C}^*$.

The third one follows from the fact that $\mathbb{C}^*$ is *divisible* while $\mathbb{Z}$ is not.

We now characterize all discrete valuations on $\mathscr{M}(X)$.

Theorem 2.4.8. *Let M be a compact Riemann surface, v a discrete valuation on $\mathscr{M}$, then there exists a unique $p \in M$ such that $v = v_p$.*

Proof. Since v is surjective we can take $h \in \mathscr{M}$ such that $v(h) = 1$. Then by properties above we have h is not constant (otherwise $v(h) = 0$), and for any $a \in \mathbb{C}^*$, $v(h - a) =$

$\min(v(h), v(a)) = 0$. Therefore for any rational function r, writing it as product of monomials we obtain

$$v(r(h)) = v_0(r).$$

Suppose that the zero points of h are $p_1, \ldots, p_n$, counting multiplicity.

For any $f \in \mathscr{M}$, viewing h as a meromorphic map to $\mathbb{CP}^1$, we know from the previous discussion that there are rational functions $r_1, \ldots, r_n$ such that

$$f^n + r_1(h)f^{n-1} + \cdots + r_n(h) = 0.$$

Thus

$$n \cdot v(f) \geq \min_{i=1,\ldots,n} \{v_0(r_i) + (n-i)v(f)\}.$$

Therefore if $v(f) < 0$, then there is an $1 \leq i \leq n$ such that $v_0(r_i) < 0$. Since $r_i(0)$ is an elementary symmetric polynomial of the values of f at $p_1, \ldots, p_n$, we then know that some of p_i must be a pole of f.

Considering $\frac{1}{f}$ we also conclude that, $v(f) > 0$ implies that one of $p_1, \ldots, p_n$ must be a zero of f.

Denote the distinct points of $p_1, \ldots, p_n$ as $q_1, \ldots, q_m$. Using Riemann-Roch theorem for divisors we can show latter the existence of a meromorphic function g such that:

- g is holomorphic at each q_i;
- $0 \neq c_i := g(q_i)$ are pairwise distinct;
- The differential $\mathrm{d}g$ does not vanish at any one of the q_i's.

Then by our discussion before, $v(g) = 0$. For the same reason we have

$$v\left(h \cdot \prod_{i=1}^{m} (g - c_i)^{-v_{q_i}(h)}\right) = 0.$$

That is,

$$1 = v(h) = \sum_{i=1}^{m} v_{q_i}(g) \cdot v(g - c_i).$$

Since $v(g - c_i) \geq \min(v(g), c_i) = 0$, there exists a unique k such that

$$v(g - c_k) = 1 \ , \ v(g - c_j) = 0 \ , \ \forall \ j \neq k.$$

Now for any $f \in \mathscr{M}$, we have

$$v\left(f \cdot \prod_{i=1}^{m} (g - c_i)^{-v_{q_i}(f)}\right) = 0,$$

which implies that

$$v(f) = \sum_{i=1}^{m} v_{q_i}(f)v(g - c_i) = v_{q_k}(f).$$

The uniqueness part is quite obvious using Riemann-Roch theorem. □

The last theorem of this section shows that the meromorphic function field determines the information of Riemann surfaces and holomorphic maps between them.

Theorem 2.4.9. *Let M, N be Riemann surfaces, $\psi : \mathscr{M}(N) \to \mathscr{M}$ a homomorphism between field extensions of $\mathbb{C}$, then there is a unique holomorphic map $h : M \to N$ such that*

$$\phi = h^*.$$

Proof. For any $p \in M$, the map $\frac{1}{n_p} \cdot v_p \circ \phi : \mathscr{M}(N) \to \mathbb{Z}$ is a discrete valuation, here

$$n_p := \min_{g \in \mathscr{M}(N), v_p(\phi(g)) > 0} v_p(\phi(g)).$$

By the previous theorem we know that there is a unique $q \in N$ such that

$$v_q = \frac{1}{n_p} v_p \circ \phi.$$

Let $f : M \to N$, $h(p) = q$ as defined above. We verify below that h is the holomorphic map we wanted.

First we check that for any $f \in \mathscr{M}(N), p \in M$,

$$\phi(f)(p) = f(h(p)).$$

For any $c \in \mathbb{C}$, we have

$$\begin{aligned} f(h(p)) = c &\Leftrightarrow v_{h(p)}(f - c) > 0 \\ &\Leftrightarrow v_p(\phi(f - c)) > 0 \\ &\Leftrightarrow v_p(\phi(f) - c) > 0 \\ &\Leftrightarrow \phi(f)(p) = c. \end{aligned}$$

Since $f(h(p)) = \infty \Leftrightarrow \frac{1}{f}(h(p)) = 0$ and ϕ is a field homomorphism, we conclude that the meromorphic functions $f \circ h$ and $\phi(f)$ are identical.

Now we prove that h is continuous: otherwise there exists a sequence $p_n \to p \in M$ such that $h(p_n) \to q \neq q_0 = h(p) \in N$. For any $f \in \mathscr{M}(N)$, we have

$$f(q) = \lim_{n \to \infty} f(h(p_n)) = \lim_{n \to \infty} \phi(f)(p_n) = \phi(f)(p) = f(h(p)) = f(q_0).$$

This means that $\mathscr{M}(N)$ cannot distinguish $q \neq q_0 \in N$, which is again impossible by Riemann-Roch theorem.

Finally we prove that h is holomorphic. For any point $p \in M$, take a meromorphic function f that is biholomorphic in a neighborhood U of $h(p)$. Since h is continuous, we can take a neighborhood V of p such that $h(V) \subset U$ and then we have $\phi(f)(V) = f \circ h(V) \subset f(U)$.

Since $\phi(f) = f \circ h$ on V and f is invertible on U, we have

$$h = f|_U^{-1} \circ \phi(f)$$

is holomorphic on V. □

Corollary 2.4.10. *If ϕ is a field isomorphism, h is a biholomorphism.*

2.5 Uniformization

In this section we prove the following theorem, which must be amazing (and unexpected) at the first glance, for it seems too strong to hold (at least to me).

Theorem 2.5.1 (Unifomization Theorem)**.** *Let X be a connected and simply connected Riemann surface. Then X is equivalent to $\mathbb{CP}^1, \mathbb{C}$ or the upper half plane $\mathbb{H}$.*

As a corollary, combining with theorem 2.1.25, we obtain:

Corollary 2.5.2. *Any Riemann surface is equivalent to one of the following:*

- $\mathbb{CP}^1$;
- $\mathbb{C}$, $\mathbb{C}^* = \mathbb{C} - \{0\}$, *or* $\mathbb{C}/\Lambda$ *for some lattice* $\Lambda \subset \mathbb{C}$;
- *A quotient* $\mathbb{H}/\Gamma$, *where* $\Gamma \subset PSL(2, R)$ *is a discrete subgroup acting freely on* $\mathbb{H}$.

To prove this theorem, of course we need ways to construct global meromorphic functions on Riemann surfaces, the following theorem is then essential for us to do this.

Theorem 2.5.3 ("Main Theorem")**.** *Let X be a compact connected Riemann surface and let ρ be a 2-form on X. There is a solution f to the equation $\nabla^2 f = \rho$ if and only if the integral of ρ over X is zero, and the solution is unique up to the addition of a constant.*

We will actually use a slightly different version of "the main theorem" in the proof of the uniformization theorem. Before stating it we give some basic definitions:

Definition 2.5.4. If ϕ is real-valued function on a non-compact space X and c is a real number, we say that ϕ tends to c at infinity in X if for all $\epsilon > 0$ there is a compact subset K of X such that $|\phi(x) - c| < \epsilon$ if x is not in K.

We say that ϕ tends to $+\infty$ at infinity in X if, for all $A \in \mathbb{R}$, there is a compact set $K \subset X$ such that $\phi(x) > A$ if x is not in K (and we say that ϕ tends to $-\infty$ at infinity in X if $-\phi$ tends to $+\infty$).

Theorem 2.5.5 ("Main Theorem*")**.** *Let X be a connected and simply connected Riemann surface. Then if ρ is a (real) 2-form of compact support on X with $\int_X \rho = 0$, there is a (real-valued) function ϕ on X with $\nabla^2 \phi = \rho$ and such that ϕ tends to 0 at infinity in X.*

We assume that theorem 2.5.3 and theorem 2.5.5 are true in this section, and leave the proofs to the next section. And now we prove the uniformization theorem.

Proof. Choose a point $p \in X$ and a local complex coordinate z around p. Let β be a bump function which is supported in the chart domain of z and constantly 1 in a smaller disc centered at p. Define

$$A := \bar{\partial}\left(\beta \frac{1}{z}\right),$$

A is a (0, 1)-form supported in an annulus around p. We put $\rho = \partial A$, so ρ is a complex-valued 2-form with integral zero by Stokes' Theorem, since d = ∂ for (0,1)-forms.

By 2.5.5 (applied to the real and imaginary parts of ρ), we can find a complex-valued function g with $\partial\bar{\partial}g = \rho$, and with the real and imaginary parts of g tending to 0 at infinity in X. Now let a be the real 1-form given by

$$a = (A - \bar{\partial}g) + \overline{(A - \bar{\partial}g)}.$$

By construction we know that $\mathrm{d}(A - \bar{\partial}g) = \partial(A - \bar{\partial}g) = 0$, so $\mathrm{d}a = 0$.

Since X is simply connected, $H^1(X) = 0$, therefore there exists a real-valued function ψ such that $\mathrm{d}\psi = a$, which means that $A - \bar{\partial}g = \bar{\partial}\psi$, i.e. $A = \bar{\partial}g + \bar{\partial}\psi$, or

$$\bar{\partial}\left(\beta\frac{1}{z} - g - \psi\right) = 0.$$

Thus $f = \beta\dfrac{1}{z} - (g + \psi)$ is a meromorphic function on X with a single pole p. The imaginary part of f tends to zero at infinity in X, since ψ is real. (Strictly, the imaginary part of f is not a function on X, since f has a pole, but the meaning should be clear to be precise, we could say that the imaginary part of f tends to 0 at infinity on $X - D$, where D is an open disc about p.)

The meromorphic function f is a holomorphic map from X to $\mathbb{CP}^1$. Let $\mathbb{H}_+$ and $\mathbb{H}_-$ denote the (open) upper and lower halfplanes in $\mathbb{C}$. Let $X_\pm$ be the preimages $f^{-1}(\mathbb{H}_\pm)$ in X. So X_+ and X_- are open subsets of X, and $X_+ \cup X_-$ is dense in X, since f is an open map. The restrictions of f give holomorphic maps

$$f_\pm : X_\pm \to \mathbb{H}_\pm.$$

We claim that f_+ and f_- are proper maps. For if B is a compact subset of $\mathbb{H}_+$, say, there is an $\epsilon > 0$ such that $Im(z) > \epsilon$ for all z in B. The fact that the imaginary part of f tends to zero at infinity implies that $f^{-1}(B)$ is a compact subset of X by definition, but this is the same as $f_+^{-1}(B) \subset X_+$.

Thus the degrees $d_\pm$ of $f_\pm$ are well-defined. And since f maps as a local homeomorphism near the pole p, we know that $X_\pm$ are both non-empty, thus $d_\pm \geq 1$. We claim that $d_\pm = 1$. To see this, we apply the condition that $Im(f)$ tends to zero at infinity to find a compact set K in X such that $Im(f)(x) < 1$ if x is not in K. Suppose the degree of f_+ is at least 2. Then, for each integer $n \geq 1$, we can find a pair of points $x_n, y_n \in X_+$ such that $f(x_n) = f(y_n) = in$ and either x_n, y_n are distinct or $x_n = y_n$ and the derivative of f vanishes at x_n. The choice of K means that x_n, y_n lie in this compact set, so we can find a subsequence $\{n'\}$ such that $x_{n'}$ and $y_{n'}$ converge to limits x, y. Since the points in, regarded as points of $\mathbb{CP}^1$, converge to ∞, we must have $f(x) = f(y) = \infty$, and since f has just one pole, we must have $x = y = p$. But now we get a contradiction to the fact that f is a local homeomorphism, with non-vanishing derivative, on a neighbourhood of p.

So now we know that f maps $X_{\pm}$ bijectively to $\mathbb{H}_{\pm}$. We claim next that f is an injection from X to $\mathbb{CP}^1$. For if x_1, x_2 are distinct points of X with $f(x_1) = f(x_2) = z \in \mathbb{CP}^1$, we can find disjoint open discs D_1, D_2 about x_1, x_2 and a neighbourhood N of $z \in \mathbb{CP}^1$ such that $f(D_1), f(D_2)$ both contain N. W.l.o.g. we assume that $N \cap \mathbb{H}_+ \neq \varnothing$. Now pick a point z' in $N \cap \mathbb{H}_+$. There are distinct points x_1', x_2' in D_1, D_2 which map to z', and this contradicts the fact that f is injective on X+.

Now we know that f maps X injectively to an open subset U of $\mathbb{CP}^1$, containing $\mathbb{H}_+ \cup \mathbb{H}_- \cup \{\infty\}$. That is,

$$U = \mathbb{CP}^1 - I,$$

for some compact subset I of R. Thus f yields an equivalence between X and this subset U. If I has more than one component, then $\pi_1(U)$ is non-trivial, which would contradict the fact that X is simply connected. Thus one of the followings hold:

- $I = \varnothing$, then X is equivalent to $\mathbb{CP}^1$;
- I is a single point, then X is equivalent to $\mathbb{C}$;
- I is a closed interval, then X is equivalent to $\mathbb{H}$, we leave it as an easy exercise to give the biholomorphic map from $\mathbb{CP}^1 - I$ to $\mathbb{H}$.

□

2.6 Proof of "the main theorem"

Well, at least it is easy to prove assuming the Hodge decomposition theorem.

Chapter 3

Compact Riemann surfaces

In this chapter we develop the elegant theory of compact Riemann surfaces.

3.1 Vector bundles on Riemann surfaces

Definition 3.1.1. Let M be a (real or complex) manifold, a **complex (real) vector bundle** $E \xrightarrow{\pi} M$ consists of a (real or complex) manifold E and a *smooth* map $\pi : E \to M$ such that there exists an open covering $\{U_\alpha\}_{\alpha \in I}$ of M and *diffeomorphisms* $\psi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}^r$ (or $\mathbb{R}^r$ if it is a real vector bundle) satisfying the following:

- The following diagram commutes:

$$\begin{array}{ccc} \pi^{-1}(U_\alpha) & \xrightarrow{\psi_\alpha} & U_\alpha \times \mathbb{C}^r \\ & \searrow^{\pi} & \downarrow{\scriptstyle pr_1} \\ & & U_\alpha \end{array}$$

  where pr_1 is the natural projection to the first component.

- For $\alpha, \beta \in I$ and $x \in U_\alpha \cap U_\beta$, the map $(pr_2 \circ \psi_\beta|_{E_x}) \circ (pr_2 \circ \psi_\alpha|_{E_x})^{-1} : \mathbb{C}^r \to \mathbb{C}^r$ (or $\mathbb{R}^r \to \mathbb{R}^r$) is a (complex or real) linear map, or equivalently the fibers $E_x = \pi^{-1}(x)$ are vector spaces and the $pr_2 \circ \psi_\alpha$'s are linear isomorphisms.

In this case r is called the **rank** of the vector bundle, and the ψ_α's are called the **local trivializations** or **bundle charts** of E. A vector bundle of rank 1 is called a **line bundle**.

If we replace the words "smooth" and "diffeomorphisms" by "holomorphic" and "biholomophirsms", we will obtain the definition of a **holomorphic vector bundle**.

Example. *We have the* ***trivial bundles*** $\underline{\mathbb{R}}^r$ *and* $\underline{\mathbb{C}}^r$ *defined as direct products of the base space M. This is also how the name "trivialization" comes.*

Now Let $g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL_r(\mathbb{C})$ be defined as

$$g_{\alpha\beta}(x) := (pr_2 \circ \psi_\alpha|_{E_x}) \circ (pr_2 \circ \psi_\beta|_{E_x})^{-1},$$

i.e.

$$\psi_\alpha \circ \psi_\beta^{-1} : U_\alpha \cap U_\beta \times \mathbb{C}^r \to U_\alpha \cap U_\beta \times \mathbb{C}^r, (x, v) \mapsto (x, g_{\alpha\beta}(x)v),$$

form which we can see that the maps $g_{\alpha\beta}$ are smooth (*holomorphic* if the vector bundle is holomorphic) and satisfies the following **cocycle condition**:

$$g_{\alpha\beta} \cdot g_{\beta\gamma} \cdot g_{\gamma\alpha} \equiv id \quad \text{on } U_\alpha \cap U_\beta \cap U_\gamma.$$

Note that this implies

$$g_{\alpha\alpha} \equiv id$$

and

$$g_{\beta\alpha} = g_{\alpha\beta}^{-1}.$$

We claim that the **transition functions** $g_{\alpha\beta}$ actually contains all the information of the vector bundle.

Given an open cover $\{U_\alpha\}_{\alpha\in I}$ of M and a class of smooth (holomorphic) maps $g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL_r(\mathbb{C})$ satisfying the cocycle condition, we set

$$E = \left(\bigsqcup_{\alpha\in I} U_\alpha \times \mathbb{C}^r \right) \Big/ \sim,$$

where " $\equiv$ " is defined as follows: for $(p, u) \in U_\alpha \times \mathbb{C}^r$ and $(q, v) \in U_\beta$,

$$(p, u) \sim (q, v) \Leftrightarrow p = q, u = g_{\alpha\beta}(p)v.$$

Then the projection map $\pi : E \to X$ is then well defined. Note that the cocycle conditions ensures that " $\sim$ " is an equivalence relation. Then one can check that $E \xrightarrow{\pi} M$ gives a (holomorphic) vector bundle whose transition functions are exactly $g_{\alpha\beta}$, and that any other vector bundle with these transition functions is isomorphic to this bundle.

Given some vector bundles we can do some simple operations between them. Suppose that E_1 and E_2 are vector bundles over M, then we can define the following vector bundles by doing the operations fiber-wisely and do these operations to the local trivializations

$$E_1 \oplus E_2, E_1 \otimes E_2, E_1 \wedge E_2, E_1^*, E_1/E_2, \ldots$$

where to define E_1/E_2 we must require that E_2 is a **subbundle** of E_1,i.e., E_2 is a submanifold of E_1 such that there exists local trivializations of E_1 under which the fibers of E_2 are just the vectors whose last several components are 0 (these are called the **subbundle charts**).

Suppose that the transition maps of E_1 and E_2 are $g_{\alpha\beta}$ and $g'_{\alpha\beta}$, then the transition maps of these vector bundles are

$$g_{\alpha\beta} \oplus g'_{\alpha\beta} = \begin{bmatrix} g_{\alpha\beta} & 0 \\ 0 & g'_{\alpha\beta} \end{bmatrix}, g_{\alpha\beta} \otimes g'_{\alpha\beta}, g_{\alpha\beta} \wedge g'_{\alpha\beta}, g^*_{\alpha\beta}, \tilde{g}_{\alpha\beta},$$

where for the last one, under the subbundle charts we have

$$g_{\alpha\beta} = \begin{bmatrix} g'_{\alpha\beta} & * \\ 0 & \tilde{g}_{\alpha\beta} \end{bmatrix}.$$

Example. *Let X be a Riemann surface, for any point $p \in X$ and $k \in \mathbb{Z}$, let (U_1, z) be a chart around the point p such that $z(p) = 0$, and let $U_2 := X - \{p\}$, we set*

$$g_{12} : U_1 \cap U_2 \to \mathbb{C}^* = GL_1(\mathbb{C}), g_{12}(q) = z(q)^k$$

on $U_1 \cap U_2 = U_1 - \{p\}$, then this gives a line bundle denoted by $L(kp)$, called a ***point bundle*** *at p.*

Let $D = \sum_{i=1}^m k_i p_i$ be a ***divisor*** *on a compact surface X,i.e. the $\mathbb{Z}$-coeffitient combinations of some finite points, then we define the* ***line bundle associated to the divisor*** *D to be*

$$L(D) := L(k_1 p_1) \otimes L(k_2 p_2) \otimes \cdots \otimes L(k_m p_m).$$

Note that these line bundles are all holomorphic. *Line bundles obtained in this way are the most important examples of line bundles over Riemann surfaces, and it turns out that every holomorphic line bundle over a compact Riemann surface arises in this way. We will use this in later sections without giving a proof. See, for example, [Don11] for a proof of this.*

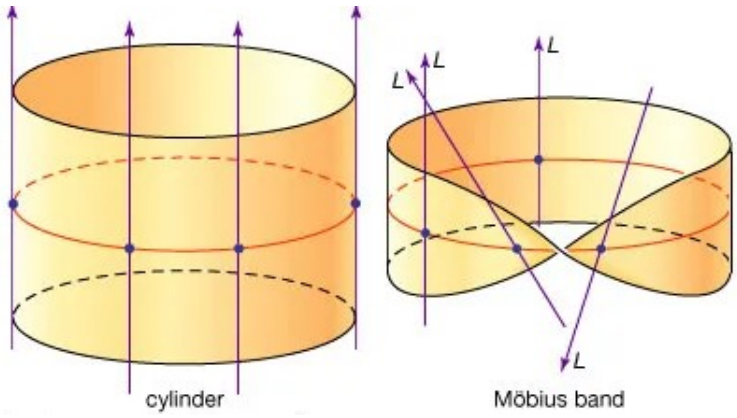


Figure 3.1: Two different line bundles over S^1, one sees that the difference is about how the bundle "twists", as indicated by the construction of $L(D)$

Suppose that $f : M \to N$ is a smooth map and $E \xrightarrow{\pi} N$ is a vector bundle, we can plant the fibers of E on M via f to obtain a new vector bundle, called the **pullback** of E:

$$f^*E := \{(p, v) : p \in M, v \in E_{f(p)}\} \subset M \times E$$

then we have the following commutative diagram:

$$\begin{array}{ccc} f^*E & \xrightarrow{F} & E \\ \downarrow{\scriptstyle f^*\pi} & & \downarrow{\scriptstyle \pi} \\ M & \xrightarrow{f} & N \end{array}$$

where F is just the projection to the second coordinate.

In terms of transition functions, the pullback bundle is just the vector bundle with transition maps $g_{\alpha\beta} \circ f$.

As one can imagine from the definition, the pullback is obtained by planting fibers of the bundle $E \xrightarrow{\pi} N$ on M via f.

Another class of examples of vector bundles arise as tensors of TM and T^*M, and on Riemann surface X we have the bundles given by

$$\Gamma(X, K) = \Omega_X^{1,0} \text{ and } \Gamma(X, \overline{K}) = \Omega_X^{0,1},$$

they are called the **canonical bundle** and the **anti-canonical bundle** of X and are naturally isomorphic to the cotangent bundle as real vector bundles.

One can notice that K is a holomorphic line bundle since its local frames are of the form $\mathrm{d}z$.

3.2 The degree of a line (vector) bundle

In this section we define the **degree** of a line (vector) bundle over a Riemann surface and use this number to classify all the vector bundles over a Riemann surface in the C^∞ sense.

Definition 3.2.1. Let $E \xrightarrow{\pi} M$ be a vector bundle, a **section** of this bundle is a smooth map $s : M \to E$ such that $\pi \circ s = id_M$, i.e. the following diagram commutes:

$$\begin{array}{ccc} M & \xrightarrow{s} & E \\ & \searrow^{id} & \downarrow{\scriptstyle \pi} \\ & & M \end{array}$$

or equivalently using the transition maps, a section is a class of smooth maps $\{s_\alpha : U_\alpha \to \mathbb{C}^r\}_{\alpha \in I}$ such that $s_\alpha = g_{\alpha\beta} s_\beta$ (we have $s_\alpha = (pr_2 \circ \psi_\alpha) \circ s$ in this setting). If E is holomorphic, this section is **holomorphic** if the map s is holomorphic or equivalently the maps s_α are holomorphic.

We are particularly interested in the zeros of sections of a vector bundle: one can prove that as a vector bundle the Möbius stripe has the property that any section of it has a zero as an easy exercise.

Definition 3.2.2. Let $f : M \to \tilde{N}$ be a smooth map and $N \subset \tilde{N}$ be a submanifold. Then

- f intersects N **transversally** if

$$\mathrm{d}f_p(T_pM) + T_{f(p)} = T_{f(p)}\tilde{N}, \forall p \in f^{-1}(N).$$

- If M is a submanifold of $\tilde{N}$, M and N are **transverse** if the inclusion $M \to \tilde{N}$ intersects N transversally.

- Two sections $\psi, \psi' \in \Gamma(E)$, where E is a vector bundle over M, are **transverse**, if for all $q = \psi(p) = \psi'(p), p \in M$,

$$\mathrm{d}\psi_p(T_pM) + \mathrm{d}\psi'_p(T_pM) = T_qE$$

  These definitions are satisfied particularly when the corresponding intersections are $\varnothing$.

We quote from differential topology that intersecting transversely is rather generic.

Theorem 3.2.3. *Given a section $\psi \in \Gamma(E)$, where E is a vector bundle over M, then any $\phi \in \Gamma(E)$ can be disturbed such that ψ and ϕ are transverse.*

Proof. See Lee... □

Corollary 3.2.4. *Let M be a compact surface and $L \to M$ a complex line bundle. If $\psi \in \Gamma(L)$ intersects the zero section transversally, the the zero set of ψ is finite.*

Proof. By dimensional reason, at zeros of ψ, ψ must be a function $f : U \to \mathbb{C}$ whose tangent map $\mathrm{d}f$ is an isomorphism at the point under a trivialization. Thus the statement is just a corollary of the inverse function theorem. □

Corollary 3.2.5. *If $E \to M$ is a complex vector bundle of rank ≥ 2 over a surface, then there exists a non-vanishing section of it.*

Proof. This is because of dimensional reason again. □

Corollary 3.2.6. *If $E \to M$ is a complex vector bundle of rank$E \geq 2$ over a surface M, then there exists a line bundle L such that*

$$E \cong_{C^\infty} \underline{\mathbb{C}}^{r-1} \oplus L.$$

Proof. Let ψ be a nowhere vanishing section of E. Define a line bundle L' to be the span of ψ, which is isomorphic to $\underline{\mathbb{C}}$.

Choose a Hermitian form on E (which can be constructed via a partition of unity) and define E' to be the orthogonal complement of L' w.r.t. this Hermitian form. Then $E = L' \oplus E'$ and E' is a complex vector space of rank $r - 1$.

Repeat this process as lone as we still have $rank(E) \geq 2$ and finally we obtain the result. □

Corollary 3.2.7. *Continuing with the last corollary, we have* $L \cong \bigwedge^r E$.

Proof. Let $\psi_1, \ldots, \psi_{r-1}$ be globally independent sections framing $\underline{\mathbb{C}}^{r-1}$. Define a map $L \to \bigwedge^r E$ by

$$v \in L_p \mapsto \psi_1(p) \wedge \cdots \wedge \psi_{r-1}(p) \wedge v \in \Lambda^r E.$$

This defines an isomorphism. □

Thus we can see that *in the* C^∞ *sense*, the task of classifying all vector bundles over a surface only requires a classification of line bundles, which will be done in this section.

We show in the following that, in the C^∞ sense, the space of line bundles over a surface X looks just like $\mathbb{Z}$.

In the following we always consider compact surfaces.

Definition 3.2.8. Let $L \to X$ be a complex line bundle and ψ a section that is either transverse to the zero section or holomorphic (if L is holomorphic), we define

$$\deg(L) := \begin{cases} \sum_{\psi(p)=0} \operatorname{sign}_p(\psi) & \text{if } \psi \text{ is transversal ;} \\ \sum_{\psi(p)=0} \operatorname{ord}_p(\psi) & \text{if } \psi \text{ is holomorphic.} \end{cases}$$

This definition seems natural if it is named by "the degree of a section of a line bundle" rather than "the degree of a line bundle", what we will show in the rest of this section is that the degree is really well-defined, that is, it does not depend on the choice of the section.

To begin with let's define the concept of a (complex) connection on a vector bundle.

Definition 3.2.9. Let $E \xrightarrow{\pi} M$ be a vector bundle, a **connection** ∇ on E is a $\mathbb{C}$-linear mapping

$$\nabla : \Gamma(M, E) \to \Gamma(M, T^*M \otimes E)$$

satisfying the Leibniz rule:

$$\nabla(f \cdot s) = \mathrm{d}f \otimes s + f \cdot \nabla s, \quad \forall\, f \in C^\infty(M)$$

The value $\nabla s(X) \in \Gamma(M, E)$ is denoted by $\nabla_X s$.

In the case E is a *holomorphic* line bundle, the connection is called **complex** if it sends holomorphic (local) sections to (local) homomorphic sections. For the existence of such connections one may refer to books on complex geometry or just let me explain for you in person.

Remark. To be explicit, ∇ is a sheaf homomorphism rather than just one linear map.

Definition 3.2.10. Let $E \xrightarrow{\pi} M$ be a vector bundle on an manifold, and ∇ a connection on it, the **curvature endomorphism** $\mathscr{R}^\nabla \in \Gamma(M, T^*M \otimes T^*M \otimes E^* \otimes E)$ is defined by

$$\mathscr{R}(X, Y, s) := \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s.$$

We define for a vector bundle E on a surface the **degree** of it by

$$\deg' E := \frac{i}{2\pi}\int_M \operatorname{tr}\left(\mathscr{R}^\nabla\right),$$

here trace is taken in $E^*\otimes E$, in particular for line bundles $E^*\otimes E\equiv\underline{\mathbb{C}}$ canonically by the trace map.

One might doubt again whether $\deg'$ depends on the choice of the connection on the vector bundle. We will answer this question for line bundles in this text via the following theorem:

Theorem 3.2.11. *The two ways for defining the degree of a line bundle agree, and therefore the first one does not depend on the choice of the section and the second one does not depend on the choice of the connection. Here when E is holomorphic we ask for the connection to be complex.*

Proof. Locally, around every point $p\in X$ we can find frame $\phi\in\Gamma(U,E|_U)$ of the line bundle E, thus since $\nabla\phi\in\Gamma(U,T^*M|_U\otimes E|_U)$ we have

$$\nabla\phi=\omega\otimes\phi$$

for a unique 1-form defined on U, called the **connection 1-form w.r.t. the section** ϕ of ∇ on U

In this case, for any $X,Y\in\Gamma(U,TM|_U)$, we have

$$\begin{aligned}\mathscr{R}(X,Y)(\phi)&=\nabla_X\nabla_Y\phi-\nabla_Y\nabla_X\phi-\nabla_{[X,Y]}\phi\\&=\nabla_X(\omega(Y)\cdot\phi)-\nabla_Y(\omega(X)\cdot\phi)-\omega([X,Y])\cdot\phi\\&=(X(\omega(Y))\cdot\phi+\omega(Y)\nabla_X\phi)-(Y(\omega(X))\cdot\phi+\omega(X)\cdot\nabla_X\phi)-\omega([X,Y])\cdot\phi\\&=(X(\omega(Y))-Y(\omega(X))-\omega([X,Y]))\cdot\phi+(\omega(X)\omega(Y)-\omega(Y)\omega(X))\phi\\&=\mathrm{d}\omega(X,Y)\cdot\phi,\end{aligned}$$

i.e.

$$\mathscr{R}=\mathrm{d}\omega.$$

Notice that the last line above depends on the fact that E is a *line bundle*, so $\omega(X)$ and $\omega(Y)$ naturally commutes. In general ω is a 1-form with matrix coefficients, and by the same computation we conclude that $\mathscr{R}=\mathrm{d}\omega+\omega\wedge\omega$.

Now let ϕ be a transverse section of E, denote by $p_1,\dots,p_n$ its distinct zeros, pick disjoint disc neighborhoods $U_1,\dots,U_n$ of them respectively such that we have frames ϕ_i on U_i respectively. Let ω_i be the connection 1-form w.r.t. the frame ϕ_i on U_i. Also let ω be the connection 1-form w.r.t the section ψ on the open set $U:=X-\{p_1,\dots,p_n\}$.

Then on each U_i, suppose that $\psi=g_i\cdot\phi_i$ then from the definition of the connection 1-forms we have

$$g_i\omega\otimes\phi_i=\omega\otimes\psi=\nabla(\psi)=\nabla(g_i\phi_i)=\mathrm{d}g_i\otimes\phi_i+g_i\nabla\phi_i=(\mathrm{d}g_i+g_i\omega_i)\otimes\phi_i,$$

which implies that $\omega = \omega_i + g_i^{-1}\mathrm{d}g_i$.

Therefore we have by Stokes' theorem that

$$\begin{aligned}\int_X \mathscr{R} &= \int_{U-\cup_i U_i} \mathscr{R} + \sum_i \int_{U_i} \mathscr{R} \\ &= \int_{U-\cup_i U_i} \mathrm{d}\omega + \sum_i \int_{U_i} \mathrm{d}\omega_i \\ &= \int_{-\cup_i \partial U_i} \omega + \sum_i \int_{\partial U_i} \omega_i \\ &= \sum_i \int_{U_i} \omega_i - \omega \\ &= -\sum_i \int_{U_i} g_i^{-1}\mathrm{d}g_i.\end{aligned} \tag{3.2.1}$$

Notice that since the section ψ is transversal, we may assume by the inverse function theorem that the map $g_i : U_i \to \mathbb{C}$ is a diffeomorphism onto its image, then by the substitution of variable formula we know that

$$\int_{U_i} g_i^{-1}\mathrm{d}g_i = \int_{g_i(\partial U_i)} \frac{\mathrm{d}z}{z} = \pm 2\pi i,$$

here the sign $\pm$ is positive iff g_i is orientation-preserving, and thus it is equal to $\mathrm{sign}_{p_i}(\psi)$, therefore

$$\frac{i}{2\pi}\int_X \mathscr{R}^\nabla = -\frac{i}{2\pi}\sum_i 2\pi i \,\mathrm{sign}_{p_i}(\psi) = \sum_{\psi(p)=0} \mathrm{sign}_p(\psi).$$

For holomorphic line bundles with complex connections and holomorphic sections respectively, the functions g_i are holomorphic in this case, and therefore the (still discrete) summation be come summation of $\frac{1}{2\pi i}\int_{\partial U_i} g_i^{-1}\mathrm{d}g_i = \mathrm{ord}_{p_i}(\psi)$. □

Remark. The proof also works for *meromorphic* sections.

Remark. Readers that are familiar with algebraic topology might have noticed that the degree is just (integral of) the Euler class of the bundle, and therefore only depends on the orientation of the bundle, which is automatic since the bundle is a complex line bundle. Also knowing that the (integral of) the Euler class of TM is $\chi(M)$, we then have proved the Gauss-Bonnet theorem here. See [BT82] for more discussions about algebraic topology.

Proposition 3.2.12.

$$\deg(E \otimes F) = \deg E + \deg F$$

Proof. Let ∇^E and ∇^F be connections on E and F, for the induced (via Leibniz rule) connection $\nabla^{E\otimes F}$, we have

$$\mathscr{R}^{\nabla^{E\otimes F}} = \mathscr{R}^{\nabla^E} \otimes id_F + id_E \otimes \mathscr{R}^{\nabla^F}.$$

□

Proposition 3.2.13.

$$\deg L(p) = 1.$$

Proof. The point bundle $L(p)$ naturally has a section s_p whose only zero is the first order zero at p. □

Proposition 3.2.14.

$$\deg K = 2g - 2$$

Proof. This is just Gauss-Bonnet theorem since as real vector bundles we have $K \equiv T^*X$. □

Form discussion above wee can see that the degree map is a group homomorphism

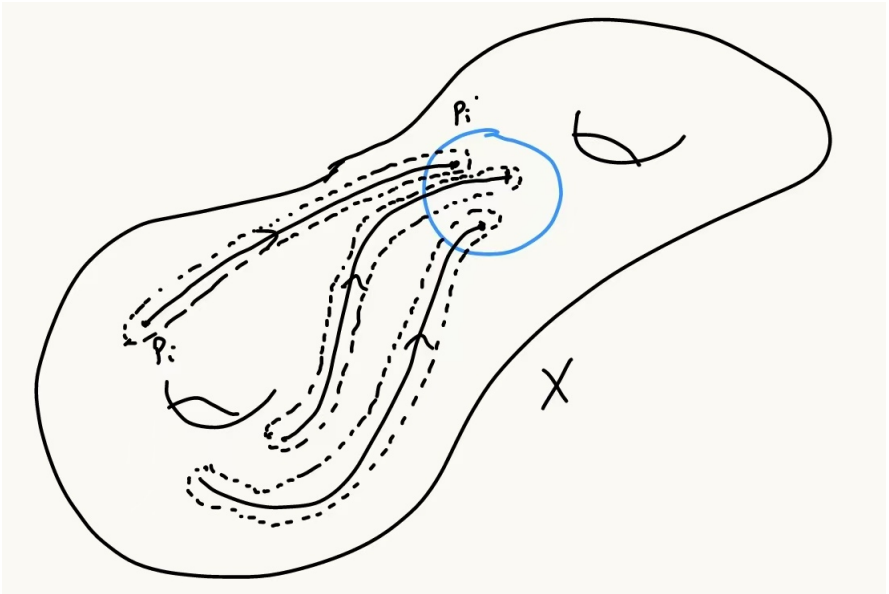
$$\deg : (\{\text{equivalent classes of line bundles over X}\}, \otimes) \to (\mathbb{Z}, +),$$

which is surjective since $\deg L(p) = 1$. We will show in the following that this is indeed an isomorphism:

Theorem 3.2.15 (C^∞ Classification of line bundles)**.** *If two complex line bundles have the same degree, the they are isomorphic in the C^∞ sense.*

Proof. We only need to prove that the kernel of the degree map is trivial. For a line bundle E with $\deg E = 0$, pick a transversal section ψ of it, and suppose $p_1, \ldots, p_n$.

Take chart discs $U \subset U_1$ and pick $p_1', \ldots, p_n'$ distinct in U, then using flow of vector bundles we can obtain a class of diffeomorphisms $G_t : X \to X : t \in [0, 1]$ such that $G_0 = id$, $G_1(p_i) = p_i'$. For each t, the bundle $E_t := E \xrightarrow{G_t^{-1} \circ \pi} X$ is still a line bundle which is equivalent to $E \xrightarrow{\pi} X$ (clearly it is equivalent to the pullback bundle $G_t^* E$, which only depends on the homotopy type of G_t, see [BT82] for details).



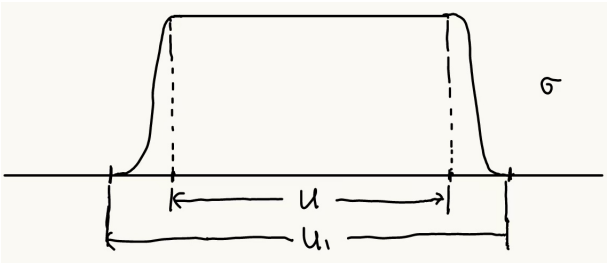


Figure 3.2: the bump function σ

Now the equivalent bundle E_1 has a section $\psi \circ G_1$, whose all zeros are $p_1', \ldots, p_n' \in U$ and has the same sign at p_i' as that of ψ at p_i.

Therefore we may assume w.l.o.g that $p_1, \ldots, p_n \in U$. Pick a local frame ϕ of E in U_1 and assume $\psi = g \cdot \phi$, then $g : U \to \mathbb{C}$. Since $\deg E = 0$ we have

$$\int_\gamma \frac{\mathrm{d}g}{g} = 0$$

for any loop γ in $U_1 \setminus U$.

Therefore $\exists\, f : U_1 \setminus \overline{U} \to \mathbb{C}$ such that $g = e^f$ on $U_1 \setminus \overline{U}$. Take a bump function $\sigma : X \to [0,1]$ such that $\sigma|_{\overline{U}} \equiv 1$ and $supp\, \sigma \subset U_1$, define

$$\tilde{\psi} := \begin{cases} \phi \cdot e^{0 \cdot \sigma + f \cdot (1-\sigma)} & \text{on } U_1 \\ \psi & \text{on } X \setminus \overline{U} \end{cases}$$

Then $\tilde{\psi}$ is well-defined and is a non-vanishing smooth section of E, making E equivalent to the trivial bundle. □

Thus we have finished the classification of line (vector) bundles over a compact surface in the C^∞ sense.

After classifying line bundles over a single surface, one may also be curious about how this classification works under pullbacks:

Theorem 3.2.16. *Let $f : M \to N$ be a holomorphic map between compact surfaces, and let $E \to N$ be a line bundle, then*

$$\deg f^* E = \deg f \cdot \deg E.$$

Proof. One can disturb a transversal section ψ of $E \to N$ a little so that its zeros are never critical values of f, then the result becomes clear since f is orientation-preserving and has $\deg f$ fibers over each regular value. □

Remark. There's an alternative proof using commutativity of the following diagram:

$$\begin{array}{ccc} H^2(N) & \xrightarrow{f^*(-)} & H^2(M) \\ \downarrow{\scriptstyle \int_N(-)} & & \downarrow{\scriptstyle \int_M(-)} \\ \mathbb{C} & \xrightarrow[(-)\cdot \deg f]{} & \mathbb{C} \end{array}$$

Let's see an application of discussions made in this section.

Theorem 3.2.17 (Hurwitz's formula)**.** *Let $f : M \to N$ be a non-constant holomorphic map between compact Riemann surfaces, then*

$$b(f) = 2(g_M - 1) + \deg f \cdot 2(1 - g_N),$$

where the ***branch order*** *$b(f)$ of f is defined by*

$$b(f) := \sum_{p \in M} (\mathrm{mult}_p(f) - 1).$$

Proof. We know that $\mathrm{d}f \in \Gamma(M, T^*M \otimes f^*TN)$. Since f is holomorphic, under the identification $K_M \equiv T^*M$ and $K_N^* \equiv TN$, $\mathrm{d}f$ is a holomorphic section of the bundle

$$K_M \otimes f^*(K_N^*).$$

Indeed, under trivialization $\mathrm{d}f$ has coordinate representation $f'(z)$.

Then by definition and properties of the degree map constructed above, we know that

$$b(f) = \deg(K_M \otimes f^*(K_N^*)) = \deg K_M + \deg f \cdot \deg K_N^* = 2(g_M - 1) + \deg f \cdot 2(1 - g_N).$$

□

3.3 Serre duality and Riemann-Roch theorem

In this section we will study in details the holomorphic sections of a holomorphic line bundle. These are of great importance because, for example, the fractions of them define *meromorphic* functions on the surface.

Recall that in chapter 1 we have defined a $\bar{\partial}$ operator on differential forms, to study holomorphic sections of a holomorphic line bundle and to state the Serre duality theorem we will need to extend this definition to sections of any holomorphic line bundle.

In the following we denote by L_1L_2 the tensor product of two line bundles, since (isomorphism classes even in the holomorphic sense of) line bundles form an abelian group. For the same reason we write L^{-1} for L^*.

Let L be a holomorphic line bundle and ψ a section of it. On any holomorphic trivialization (U, s) of L, where s is a holomorphic frame, we write

$$\psi = s \cdot \xi, \ \xi : U \to \mathbb{C}$$

And define

$$\bar{\partial}(\psi|_U) := s \cdot \bar{\partial}\xi \in \Gamma(U, \bar{K}L),$$

where $\bar{\partial}\xi$ is defined as before. This in fact gives a global section of $\bar{K}L$: Let s' be another local holomorphic frame of L, and suppose that $s = s' \cdot g$ at points both sides are defined, then g is holomorphic. So $\bar{\partial} g = 0$, and

$$s' \cdot \bar{\partial}\xi' = s' \cdot \bar{\partial}(g\xi) = s' \cdot (g\bar{\partial}\xi + \xi\bar{\partial}g) = s' \cdot g\bar{\partial}\xi = s \cdot \bar{\partial}\xi.$$

Thus we have a well defined linear map

$$\bar{\partial} : \Gamma(M, L) \to \Gamma(M, \bar{K}L)$$

satisfying

$$\bar{\partial}(fs) = \bar{\partial}f \cdot s + f \cdot \bar{\partial}s, \ \ f \in C^\infty(M, \mathbb{C}), \ \ s \in \Gamma(M, L), \tag{3.3.1}$$

and also that

$$\left(\bar{\partial}s = 0\right) \Leftrightarrow (s \text{ is holomorphic}). \tag{3.3.2}$$

An operator satisfying eq. (3.3.1) is called a **holomorphic structure**. Any two holomorphic structures differ by an element of $\Gamma(M, \bar{K}LL^*)$. And in fact, any complex vector bundle with a holomorphic structure can be turned into a holomorphic line bundle in a unique way satisfying eq. (3.3.2). One may turn to books about complex geometry to see detailed discussions.

Definition 3.3.1. Let L be a line bundle over a Riemann surface M with holomorphic structure $\bar{\partial}$, define the **Dolbeault cohomology groups**

$$H^0_{\bar{\partial}}(E) := \ker \bar{\partial}|_{\Gamma(E)},$$

and

$$H^1_{\bar{\partial}}(E) := \Gamma(\bar{K}E)/\bar{\partial}\Gamma(E).$$

Notice that H^0 consists of (global) holomorphic sections exactly.

Before stating the Serre duality theorem, let's first observe the following lemma

Lemma 3.3.2.

$$\bar{K}K \cong K\bar{K} \cong \Omega^2$$

Proof. The isomorphism is given by $f \otimes g \mapsto f \wedge g$. □

We will not distinguish these two bundles in the future.

Now we can state the Serre duality theorem, which is kind of a "complex Poincaré" duality theorem:

Theorem 3.3.3 (Serre duality theorem)**.** *For a holomorphic line bundle L over a compact Riemann surface M, we have a canonical isomorphism*

$$H^0(L) \cong (H^1(KL^{-1}))^*$$

give by the pairing

$$H^0(L) \otimes H^1(KL^{-1}) \to \mathbb{C}$$
$$\phi \otimes \psi \mapsto \int_M \langle \phi, \psi \rangle = \int_M \psi(\phi).$$

We will not prove the theorem for general holomorphic line bundles, instead we will prove for line bundles that comes form divisors (which are actually all holomorphic line bundles, but we may not prove this, either.)

The reason we want to deal with divisors is that they have "inductive" nature so that we can prove via adding points one by one and using induction.

Lemma 3.3.4. *For any holomorphic line bundle $L \to M$, $\dim_{\mathbb{C}} H^0(L) < \infty$.*

Proof. Let d be the degree of L, we prove below that $\dim_{\mathbb{C}} H^0(L) \leq d+1$.

Otherwise let $s_1, \ldots, s_{d+2}$ be linearly independent holomorphic sections of the bundle, in any local frame (U, σ) near an arbitrary point p, we assume that $s_i = f_i \cdot \sigma$, then the system of linear equations

$$\begin{aligned}
&\sum_{i=1}^{d+2} c_i \cdot g_i(p) = 0 \\
&\sum_{i=1}^{d+2} c_i \cdot g_i'(p) = 0 \\
&\cdots \\
&\sum_{i=1}^{d+2} c_i \cdot g_i^{(d)}(p) = 0
\end{aligned}$$

will have a nonzero solution $(c_1, \ldots, c_{d+2}) \in \mathbb{C}^{d+2}$, now let

$$s := \sum_{i=1}^{d+2} c_i \cdot s_i \in H^0(L),$$

then $s \neq 0$, and $\mathrm{ord}_p(s) \geq d+1$, which is impossible by definition of the degree of a line bundle. □

Lemma 3.3.5 ($\bar{\partial}$-lemma)**.** *Let $\psi \in C^1(\mathbb{C})$ with compact support. Define the integral operator:*

$$u(z) := -\frac{1}{2\pi i} \iint_{\mathbb{C}} \frac{\psi(\zeta)}{\zeta - z} d\zeta \wedge d\bar{\zeta}$$

Then the following hold:

(i) Regularity: $u \in C^1(\mathbb{C})$

(ii) Solution property: $\dfrac{\partial u}{\partial \bar{z}} = \psi$ on $\mathbb{C}$

Proof. See [GG07] □

Now let's prove the Serre duality theorem.

Proof. (I) For $L = \underline{\mathbb{C}}$, we have

$$H^0(\underline{\mathbb{C}}) \cong \mathbb{C},$$

and

$$H^1(K\underline{\mathbb{C}}^{-1}) = \frac{\Gamma(\overline{K}K)}{\bar{\partial}\Gamma(\bar{K})}.$$

What we need to prove is that the map

$$\int_M (-) : \Gamma(\overline{K}K) = \Omega^2(M) \to \mathbb{C}$$

is surjective with kernel exactly $\bar{\partial}\Gamma(\bar{K})$.

- Surjectivity: The integral of a volume form (which exists since Riemann surfaces are oriented) is positive.
- $\bar{\partial}\Gamma(\bar{K})$ is contained in the kernel: notice that on $\Gamma(K)$ we have $\mathrm{d} = \partial + \bar{\partial} = \bar{\partial}$, then this follow from Stokes' theorem.
- If $\rho \in \Omega^2(M)$, $\int_M \rho = 0$, then by theorem 2.5.3 ("the main theorem") we know that $\exists\, f : M \to \mathbb{C}$ such that

$$\rho = \Delta f = \bar{\partial}(2i\partial f) \in \bar{\partial}\Gamma(K).$$

Thus Serre duality theorem holds for $L = \underline{\mathbb{C}}$.

(II) For $L = K$, or in other words, for $L = K\underline{\mathbb{C}}^{-1}$, we have

$$H^1(KL^{-1}) = H^1(\overline{\mathbb{C}}) = \frac{\Gamma(\bar{K})}{\bar{\partial}C^\infty(M)}.$$

We prove the following lemma which gives Serre duality in this case as a corollary:

Lemma 3.3.6. *We have the following facts:*

1. *The map $\sigma : H^0(K) \to H^1(\underline{\mathbb{C}})$, $\alpha \mapsto \overline{\alpha}$ is an isomorphism;*
2. *The bilinear pairing (denooted by B) given above induces an isomorphism $H^0(K) \cong (H^1(\underline{\mathbb{C}}))^*$;*
3. *The map $H^0(K) \oplus H^1(\underline{\mathbb{C}}) \to H^1_{dR}(M, \mathbb{C})$ defined by*

$$(\alpha, \theta) \mapsto [\alpha + \overline{\sigma^{-1}(\theta)}]$$

is an isomorphism. In particular, we have

$$\dim_{\mathbb{C}} H^0(K) = g,$$

which only depends on the topology (rather than the complex structure) of the surface.

Proof. 1. For a holomorphic 1-form α, if $\overline{\alpha} = \bar{\partial} f$, then $\alpha = \partial \bar{f}$ is holomorphic, making

$$\Delta f = 2i\bar{\partial}\partial f = 0.$$

Since M is compact we conclude that f is constant and so $\alpha = 0$.

Conversely for any $\theta \in \Gamma(\overline{K})$, we need to find a function g on M such that the form $\theta + \bar{\partial} g$, which is of the same class in $H^1(\underline{\mathbb{C}})$ as θ, is anti-holomorphic so that the form $\alpha = \overline{\theta + \bar{\partial} g}$ is holomorphic.

The condition required is just $\partial\bar{\partial} g = -\partial\theta = -\mathrm{d}\theta$. Again by Stokes' theorem we know that the RHS has total integral 0, and therefore the result follows from the main theorem.

2. The pairing $H^0(K) \times H^0(K) \to \mathbb{C}$ given by $(\alpha, \beta) \mapsto B(\alpha, \sigma(\beta))$ is up to a constant i equal to the inner product on $H^0(K)$ defined in eq. (1.4.1), which therefore gives the result.

3. Injectivity is clear, for surjectivity, consider any closed 1-form

$$\omega = \omega' + \omega'',$$

then $\bar{\partial}\omega' = \mathrm{d}\omega' = -\mathrm{d}\omega'' = -\partial\omega''$.

We aim to find a function h such that

$$\bar{\partial}(\omega' + \partial h) = 0, \partial(\omega'' + \bar{\partial} h) = 0,$$

which is equivalent to saying that

$$\Delta h = -2i\bar{\partial}\omega' = 2i\partial\omega'',$$

which again holds because of the Stokes' theorem and the main theorem.

□

From the lemma we clearly see that the Serre duality also holds for $L = K = K\underline{\mathbb{C}}^{-1}$.
(III) Suppose that for line bundle L we have

$$H^0(L) \cong (H^1(KL^{-1}))^*, H^0(KL^{-1}) \cong (H^1(L))^*$$

via the canonical pairings, we prove that these still holds when replacing L by $LL(p)$ for any point bundle $L(p) \to M$, and vice versa.

For this we need to prepare several lemmas:

Lemma 3.3.7. *For any line bundle L over M,* $\dim H^0(KL^{-1}) \leq \dim H^1(L)$.

Proof. The spirit of this lemma is that elements of $H^0(KL^{-1})$ provide *obstructions* for elements of $\Gamma(\bar{K}L)$ to lie in the image of $\bar{\partial}$.

Let $\omega_1, \ldots, \omega_n$ be a basis of $H^0(KL^{-1})$, then we claim that (after a change of basis) there exists $\phi_1, \ldots, \phi_n \in \Gamma(\bar{K}L)$ such that

$$\int_M \omega_i(\phi_j) = \delta_{ij}.$$

Since M is a surface we know that $\bar{\partial}\phi_i = 0$ for every i. If $\sum_i c_i[\phi_i] = 0$, suppose that $\sum c_i\phi_i = \bar{\partial}f$ for $f \in \Gamma(L)$, then

$$c_i = \int_M \omega_i(\bar{\partial}f) = \int_M \bar{\partial}(\omega(f)) - \int_M (\bar{\partial}\omega)f = 0,$$

noticing that $\mathrm{d} = \bar{\partial}$ on $\Gamma(K)$. Thus $[\phi_i]$ are linearly independent in $H^1(L)$.

Now we prove the claim: let $p_1, \ldots, p_m$ be distinct points on M where $m > \deg(KL^{-1})$, and let σ_i be frames of KL^{-1} near p_i and suppose $\omega_i = g_i\sigma_i$, then by the degree argument again we can see that the matrix

$$(g_i(p_j))_{\substack{1\le i\le m\\ 1\le j\le n}}$$

has full rank, the rest is then by taking bump functions near the p_i's and some simple linear algebra. □

Now consider the maps:

$$\begin{array}{ccccc}
H^0(L) & \xrightarrow{\phi_p=(-)\otimes s_p} & H^0(LL(p)) & \qquad & H^1(L) \xrightarrow{\psi_p=(-)\otimes s_p} H^1(LL(p)) \\
\otimes & & \otimes & \xleftrightarrow[L\leftrightarrow KL^{-1}L(-p)]{} & \otimes \qquad\qquad \otimes \\
H^1(KL^{-1}) & \xleftarrow[\phi_p^*=(-)\otimes s_p]{} & H^1(KL^{-1}L(-p)) & \qquad & H^0(KL^{-1}) \xleftarrow[\psi_p^*=(-)\otimes s_p]{} H^0(KL^{-1}L(-p)) \\
\downarrow & & \downarrow & & \downarrow \qquad\qquad \downarrow \\
\mathbb{C} & & \mathbb{C} & & \mathbb{C} \qquad\qquad \mathbb{C}
\end{array}$$

Lemma 3.3.8. *$\phi_p : H^0(L) \to H^0(LL(p))$ is injective with cokernel of dimension at most 1.*

Proof. Injectivity is clear, and we have a linear injection

$$coker(\phi_p) \to \mathbb{C}, f \mapsto f(p),$$

showing the cokernel condition. □

Lemma 3.3.9. *$\psi_p : H^1(L) \to H^1(LL(p))$ is surjective with kernel of dimension at most 1.*

Proof. Pick any local holomorphic frame σ on $U \ni p$. For any section $\varphi \in \Gamma(\bar{K}L)$, write $\varphi = g \cdot \sigma$ on U. Then by $\bar{\partial}$ lemma we can find some smooth function f supported in U such that $\bar{\partial} f = g$ near p. Then we have $[\varphi] = [\varphi - \bar{\partial}(f \cdot \sigma)] = [\varphi - (\bar{\partial} f) \cdot \sigma]$, and clearly $\varphi - (\bar{\partial} f) \cdot \sigma$ lie in the image of ψ_p as it is identically zero in a neighborhood of p. This gives the surjectivity.

If $\phi_1, \phi_2 \in \ker(\psi_p)$, by using the $\bar{\partial}$-lemma again, one may assume that ϕ_i are identically zero near p. Then $\exists\, t_1, t_2 \in \Gamma(LL(p))$ not all zero such that $\phi_i \otimes s_p = \bar{\partial} t_i$, $i = 1, 2$. Then there exists $a, b \in \mathbb{C}$ such that $at_1(p) + bt_2(p) = 0$. By our choice of representatives, we know that t_1, t_2 are holomorphic near p, and so is $at_1 + bt_2$. Noticing that $at_1(p) + bt_2(p) = 0$ we know that

$$(at_1 + bt_2) \otimes s_p^{-1} \in \Gamma(L),$$

making $a[\phi_1] + b[\phi_2] = 0$. □

Lemma 3.3.10. *Assume that $\phi_p : H^0(L) \to H^0(LL(p))$ is not surjective. Then:*

1. *ψ_p is injective;*
2. *ϕ_p^* is not surjective;*
3. *ψ_p^* is surjective.*

Proof. 1. Let $\phi \in \Gamma(\bar{K}L)$ s.t. $\phi \cdot s_p = \bar{\partial} h$ for some $h \in \Gamma(LL(p))$. By using the $\bar{\partial}$-lemma again we may assume that ϕ is identically zero on some neighborhood of p. Then the section h is holomorphic near p.

   By assumption we know that $\exists\, \omega \in H^0(LL(p))$ such that $\omega(p) \neq 0$. Pick $c \in \mathbb{C}$ such that $h(p) - c \cdot \omega(p) = 0$, then again by holomorphicity we know that $(h - c \cdot \omega) \otimes s_p^{-1} \in \Gamma(L)$ satisfies that

$$\bar{\partial}\left((h - c \cdot \omega) \otimes s_p^{-1}\right) = \phi,$$

   showing that $[\phi] = 0$.

2. Let s_U be a holomorphic frame of $KL^{-1}L(-p)$ defined on some small chart domain (U, z), and let h be a bump function which is supported in U and is identically 1 near p. Now consider the section

$$\bar{\partial}\left(h \cdot \frac{1}{z}\right) \cdot s_U \in \Gamma(\bar{K}KL^{-1}L(-p)),$$

   it is not perpendicular to ω defined above: let $U_1 \subset U$ be obtained by deleting a $\epsilon-$ball B_ϵ on which h is constantly 1, then noticing that $s_U(\omega)$ lies in K we have

$$\begin{aligned}\int_M \bar{\partial}\left(h \cdot \frac{1}{z}\right) \cdot s_U(\omega) &= \int_{U_1} \bar{\partial}\left(h\frac{1}{z}\right) \cdot s_U(\omega) \\ &= \int_{\partial U_1} h\frac{1}{z} \cdot s_U(\omega) - \int_{U_1} h\frac{1}{z}\, \mathrm{d}(s_U(\omega)) \\ &= \int_{\partial B_\epsilon} \frac{1}{z} \cdot s_U(\omega),\end{aligned}$$

since s_U and ω are both holomorphic. Let $s_U(\omega_p) = c_p \cdot \mathrm{d}z \in K_p$, then $c_p \neq 0$, and

$$\lim_{\epsilon \to 0} \int_{\partial B_\epsilon} \frac{1}{z} \cdot s_U(\omega) = 2\pi i c_p \neq 0,$$

showing that

$$\bar{\partial}\left(h \cdot \frac{1}{z}\right) \cdot s_U \notin \operatorname{Im} \bar{\partial}.$$

However we have

$$\bar{\partial}\left(h \cdot \frac{1}{z}\right) \cdot s_U \cdot s_p = \bar{\partial}\left(h \frac{1}{z} s_U s_p\right) \in \operatorname{Im} \bar{\partial},$$

showing that ϕ_p^* is not injective.

3. If $\psi_p^* : H^0(KL^{-1}L(-p)) \to H^0(KL^{-1})$ is not surjective, applying the previous parts of this lemma to the bundle $KL^{-1}L(p)$, we know that the map

$$H^1(K(KL^{-1}L(-p))^{-1}L(-p)) \to H^1(K(KL^{-1}L(-p))^{-1})$$

given by tensoring with s_p is not injective, which contradicts to the first statement of this lemma.

□

Lemma 3.3.11. *ϕ_p and ψ_p^* cannot be both surjective if either L or $LL(p)$ satisfies the two Serre duality conditions.*

Proof. We first show that ψ_p and ϕ_p^* are not injective:

Otherwise let s_U be a local frame of L defined on U again. Then $[\bar{\partial}(h\frac{1}{z} \cdot s_U) \cdot s_p] = 0$ implies that $\exists \mu \in \Gamma(L)$ such that

$$\bar{\partial}\mu = \bar{\partial}(h\frac{1}{z} \cdot s_U),$$

thus

$$s_p \cdot (h\frac{1}{z}s_U - \mu) \in H^0(LL(p))$$

is a holomorphic section that is not zero at p, showing that ϕ_p is not surjective.

For ϕ_p^* just use $L \mapsto KL^{-1}L(-p)$ again.

Now in the diagram:

$$\begin{array}{ccc} H^1(L) & \xrightarrow[\psi_p(\text{dim decrease by } 1)]{} & H^1(LL(p)) \\ \Big| \cong & & \Big| \dim \geq \\ H^0(KL^{-1}) & \xleftarrow{\psi_p^*(\cong)} & H^0(KL^{-1}L(-p)) \end{array}$$

in the first line the dimension decreases by 1, while in the second line the dimension remains invariant, so if L satisfies the Serre duality theorem this would contradict to lemma 3.3.7.

Similarly if $LL(p)$ satisfies the conditions we just have $KL^{-1}L(p)$ in mind and apply the same argument to the diagram

$$\begin{array}{ccc} H^1(KL^{-1}L(-p)) & \xrightarrow[\phi_p^*(\text{dim decrease by 1})]{} & H^1(KL^{-1}) \\ \Big| \cong & & \dim \geq \Big\downarrow \\ H^0(LL(p)) & \xleftarrow{\phi_p(\cong)} & H^0(L) \end{array}$$

□

Now we can do our inductions via the diagrams

$$\begin{array}{ccc} H^0(L) & \xrightarrow{\phi_p=(-)\otimes s_p} & H^0(LL(p)) \\ \otimes & & \otimes \\ H^1(KL^{-1}) & \xleftarrow[\phi_p^*=(-)\otimes s_p]{} & H^1(KL^{-1}L(-p)) \\ \downarrow & & \downarrow \\ \mathbb{C} & & \mathbb{C} \end{array} \qquad \begin{array}{ccc} H^1(L) & \xrightarrow{\psi_p=(-)\otimes s_p} & H^1(LL(p)) \\ \otimes & & \otimes \\ H^0(KL^{-1}) & \xleftarrow[\psi_p^*=(-)\otimes s_p]{} & H^0(KL^{-1}L(-p)) \\ \downarrow & & \downarrow \\ \mathbb{C} & & \mathbb{C} \end{array}$$

On the side there is no dimension change the result follows directly. On the side there is one dimensional dimension change, the result comes from the proof of lemma 3.3.10 and the facts that

$$\bar{\partial}\left(h \cdot \frac{1}{z}\right) \cdot s_U \perp \operatorname{Im} \psi_p \quad , \quad \langle \omega, \bar{\partial}\left(h \cdot \frac{1}{z}\right) \cdot s_U \rangle \neq 0.$$

□

Remark. This proof also works out the **Riemann-Roch** theorem: in lemma 3.3.6 we have shown that $\dim H^0(K) = g$, thus we have

$$\dim H^0(\underline{\mathbb{C}}) - \dim H^0(K\underline{\mathbb{C}}^{-1}) = 1 - g + \deg(\underline{\mathbb{C}}).$$

So denoting $h^0(L) = \dim H^0(L)$, via the same induction we have proved:

Theorem 3.3.12 (Riemann-Roch)**.**

$$h^0(L) - h^0(KL^{-1}) = 1 - g + \deg L,$$

or equivalently

$$\dim H^0(L) - \dim H^1(L) = 1 - g + \deg L.$$

Or maybe writing

$$\chi(L) = 1 - g + \deg L$$

is clearer for someone.

Though it remains to show that every holomorphic line bundle comes from a divisor …

Corollary 3.3.13. *1.* $h^0(L) = 0$ *for* $\deg L < 0$*;*

2. $0 \leq h^0(L) \leq \deg L + 1$*;*

3. For $\deg L \geq g - 1$, $\deg L + 1 - g \leq h^0(L) \leq g$*;*

4. For $\deg L > 2g - 2$, $h^0(L) = \deg L + 1 - g$.

Corollary 3.3.14. *Any holomorphic line bundle has a meromorphic section.*

Proof. $h^0(L \otimes L(kp)) > 0$ for $k > g - \deg L$. □

Via the same method we can show:

Corollary 3.3.15. *For any divisor* D *of degree 0,*

$$L(D) \cong L(p_1 + \cdots + p_g - q_1 - \cdots - q_g)$$

for some $p_i, q_i \in M$*. Moreover we can fix the points* $q_i \in M$*.*

In particular for $\mathbb{CP}^1$*, holomorphic line bundles are determined by their degree.*

Remark. This corollary tells us the *moduli space* of line bundles has (complex) dimension $\leq g$, we will (?) show later that this moduli space is actually a tori $\mathbb{C}^g/\Lambda$.

Holomorphic sections (actually ratios of them) of line bundles can be used to construct meromorphic functions

Corollary 3.3.16. *Any genus 2 surface is hyperelliptic.*

Proof. Exercise. □

3.4 Cohomology of sheaves and second proofs to R-R theorem and Serre duality

3.5 Abel's theorem and the Jacobi inversion theorem

We aim to classify all holomorphic line bundles (of the same degree), to do this it suffices to classify holomorphic line bundles of degree 0.

As we have seen, such line bundles are $\underline{\mathbb{C}}$ in the C^∞ sense. So instead we can ask for the moduli space of distinct holomorphic line bundles structures on $\underline{\mathbb{C}}$.

Let $\bar{\partial}_L$ be the holomorphic structure with respect to L, then $\alpha_L := \bar{\partial}_L - \bar{\partial}_0$ is a section of $\bar{K}$, and vise versa.

If two holomorphic line bundle structures on $\underline{\mathbb{C}}$ are isomorphic, we will have $g : M \to \mathbb{C}^*$ such that it sends a (local) holomorphic section ϕ to a (local) holomorphic section $g\phi$. Thus

$$0 = \bar{\partial}_{L'}(g\phi) = \bar{\partial}_0(g)\phi + g\bar{\partial}_{L'}\phi = \bar{\partial}_0(g)\phi + g(\bar{\partial}_{L'} - \bar{\partial}_L)\phi = \bar{\partial}_0(g)\phi + g(\alpha_{L'} - \alpha_L)\phi,$$

so

$$\alpha_L = \alpha_{L'} + g^{-1}\bar{\partial}g.$$

Therefore the moduli space of holomorphic line bundles of degree zero is

$$\Gamma(\bar{K})/\{g^{-1}\bar{\partial}g : g \in C^\infty(M, \mathbb{C}^*)\}.$$

Notice that what we quotient out is a subgroup, as it is the image of a group homomorphism:

$$\frac{\bar{\partial}(fg)}{fg} = \frac{\bar{\partial}f}{f} + \frac{\bar{\partial}g}{g},$$

but it does not seem to be a linear subspace.

For any $\alpha \in \Gamma(\bar{K})$, from

$$\int_M \mathrm{d}\alpha = 0$$

and the main theorem we know that $\exists\, f \in C^\infty(M)$ such that $\partial\bar{\partial}f = \mathrm{d}\alpha$, i.e. $\tilde{\alpha} := \alpha - \bar{\partial}f$ is a closed form.

Notice that, letting $g := e^f : M \to \mathbb{C}^*$ we have $\bar{\partial}g = e^f \cdot \bar{\partial}f = g \cdot \bar{\partial}f$, i.e. $\bar{\partial}f = \dfrac{\bar{\partial}g}{g}$. Thus in the quotient we have $[\alpha] = [\tilde{\alpha}]$. And as a corollary we have the moduli space of degree-zero holomorphic line bundles is given by

$$\left\{\alpha \in \Gamma(\bar{K}) : \mathrm{d}\alpha = 0\right\} \Big/ \left\{\frac{\bar{\partial}g}{g} : g : M \to \mathbb{C}^*, \mathrm{d}\left(\frac{\bar{\partial}g}{g}\right) = 0\right\}$$

Notice that

$$\left\{\alpha \in \Gamma(\bar{K}) : \mathrm{d}\alpha = 0\right\} = \overline{H^0(K)},$$

so the moduli space of degree-zero holomorphic line bundles can also be given by

$$H^0(K) \Big/ \left\{\frac{\partial g}{g} : g \in C^\infty(M, \mathbb{C}^*)\right\} \cap H^0(K)$$

Recall that we have shown that $H^0(K) \cong \mathbb{C}^g$, we have therefore reduced our discussion to a finite dimensional object.

Now, via the Dirichlet inner product on $H^0(K)$ we have an isomorphism

$$H^0(K) \xrightarrow{\cong} \left(\overline{H^0(K)}\right)^*, \alpha \mapsto \left(\beta \mapsto \int_M \alpha \wedge \beta\right)$$

We will show that under this map the subgroup in the quotient corresponds to

$$\Gamma := \left\{2\pi i \int_\gamma (-) : \gamma \in \pi_1(M)\right\}.$$

Lemma 3.5.1. *Let γ be a loop on M, f a non-vannishing smooth function defined on a neighborhood of γ, then*

$$\frac{1}{2\pi i}\int_\gamma \frac{\mathrm{d}f}{f} \in \mathbb{Z}.$$

Proof. Lift γ to the universal covering space M, which is simply connected, and then apply the argument principle to see that $\log(f)$ is well-defined on the universal covering space. □

In the following lemma we assume knowledge of the Hodge decomposition.

Lemma 3.5.2. *For any closed curve σ on M, there exists a unique harmonic 1-form of the form $\omega_\sigma = \dfrac{\mathrm{d}f}{2\pi i f}$ such that for any closed 1-form η on M we have*

$$\int_\sigma \eta = \int_M \omega_\sigma \wedge \eta.$$

Proof. Let $\sigma : [0,1] \to M$ be a closed curve on M with $\sigma(0) = \sigma(1)$. Choose a partition of $[0,1]$:

$$0 = t_0 < t_1 < \cdots < t_n = 1$$

such that for each $j = 1, \cdots, n$, $\sigma([t_{j-1}, t_j])$ is contained in a coordinate disk D^j of M. Within D^j, there exists a coordinate function z_j satisfying $z_j(D^j) = \mathbb{D}$. We may further select $0 < r < 1$ such that

$$\sigma([t_{j-1}, t_j]) \subset D^j_r = \{p \in D^j \mid |z_j(p)| < r\}.$$

Consider the meromorphic function $\dfrac{z - z(\sigma(t_j))}{z - z(\sigma(t_{j-1}))}$ in D^j_r, which has a simple zero at $\sigma(t_j)$ and a simple pole at $\sigma(t_{j-1})$. We now smoothly extend it to the entire M. To achieve this, choose $1 > r' > r'' > r$, and let ψ_j be a smooth bump function on M satisfying:

$$\psi_j(p) = 1 \quad \forall p \in D^j_{r''}; \qquad \operatorname{supp}\psi_j \subset D^j_{r'}.$$

Additionally, the function $\log \dfrac{z - z(\sigma(t_j))}{z - z(\sigma(t_{j-1}))}$ has a well-defined single-valued branch on $D^j \setminus D^j_r$. Define $f_j : M \setminus \{\sigma(t_{j-1})\} \to \mathbb{C}$ as follows:

$$f_j(p) = \begin{cases} \exp\left[\psi_j \log \dfrac{z - z(\sigma(t_j))}{z - z(\sigma(t_{j-1}))}\right], & p \in M \setminus D^j_r, \\ \dfrac{z - z(\sigma(t_j))}{z - z(\sigma(t_{j-1}))}, & p \in D^j_{r''}. \end{cases}$$

Then the function f_j is identically 1 on $M \setminus D^j_{r'}$.

Set $f = f_1 f_2 \cdots f_n$. Then f is a smooth nowhere-vanishing function defined on all of M. Define

$$\omega' = \frac{1}{2\pi i}\frac{df}{f},$$

so that ω' is a smooth 1-form on M. Let η be a closed 1-form on M. We prove that

$$\int_\sigma \eta = \int_M \omega' \wedge \eta.$$

Indeed, choose coordinate disks B^j centered at $\sigma(t_j)$. Within each D^j, there exists a function g_j such that $\eta = dg_j$.

We have

$$\begin{aligned}
\int_M \omega' \wedge \eta &= \lim_{\epsilon\to 0+} \int_{M-\bigcup_j B^j_\epsilon} \omega' \wedge \eta = \lim_{\epsilon\to 0+} \sum_{k=1}^n \frac{1}{2\pi i} \int_{M-\bigcup_j B^j_\epsilon} \frac{\mathrm{d}f_k}{f_k} \wedge \eta \\
&= \lim_{\epsilon\to 0+} \sum_{k=1}^n \frac{1}{2\pi i} \int_{D^k - B^{k-1}_\epsilon - B^k_\epsilon} \frac{\mathrm{d}f_k}{f_k} \wedge \eta = \lim_{\epsilon\to 0+} \sum_{k=1}^n \frac{1}{2\pi i} \int_{D^k - B^{k-1}_\epsilon - B^k_\epsilon} \frac{\mathrm{d}f_k}{f_k} \wedge \mathrm{d}g_k \\
&= \lim_{\epsilon\to 0+} \sum_{k=1}^n \frac{1}{2\pi i} \int_{D^k - B^{k-1}_\epsilon - B^k_\epsilon} -\mathrm{d}\left(g_k \frac{\mathrm{d}f_k}{f_k}\right) \\
&= \lim_{\epsilon\to 0+} \sum_{k=1}^n \frac{1}{2\pi i} \int_{\partial B^{k-1}_\epsilon} g_k \frac{\mathrm{d}f_k}{f_k} + \int_{\partial B^k_\epsilon} g_k \frac{\mathrm{d}f_k}{f_k} \\
&= \sum_{k=1}^n (g_k(\sigma(t_k)) - g_k(\sigma(t_{k-1}))) = \sum_{k=1}^n \int_{\sigma|_{[t_{k-1},t_k]}} \eta = \int_\sigma \eta.
\end{aligned}$$

Now notice that

$$\mathrm{d}\left(\frac{\mathrm{d}f}{f}\right) = \left(-\frac{\mathrm{d}f}{f^2}\right) \wedge \mathrm{d}f = 0,$$

thus by the Hodge decomposition we have

$$\frac{1}{2\pi i}\frac{\mathrm{d}f}{f} = \omega' = \omega_\sigma + \mathrm{d}g$$

for some unique harmonic 1-form

$$\omega_\sigma = \frac{1}{2\pi i}\frac{\mathrm{d}\left(fe^{-2\pi i g}\right)}{fe^{-2\pi i g}}.$$

The uniqueness part is clear. □

Remark. The harmonic 1-form ω_σ is called the **dual form** of the closed curve σ. This name comes form the view point that ω_σ is a canonical element of the cohomology class dual to σ in the Poincaré duality.

By the previous lemma, we know that for any two closed curves σ and τ, the integral

$$\int_M \omega_\sigma \wedge \omega_\tau$$

is an integer, called the **intersection number** of σ and τ.

Thus we know that for any σ, in $\left(\overline{H^0(K)}\right)^*$, we have

$$2\pi i \int_\sigma (-) = \int_M \frac{\mathrm{d}f}{f} \wedge (-) = \int_M \frac{\partial f}{f} \wedge (-).$$

So the $(1,0)$- part of the harmonic 1-form ω_σ (which is therefore holomorphic), $\dfrac{\partial f}{f} = \omega'_\sigma$ behaves the same as $2\pi i \int_\sigma (-)$ on $\overline{H^0(K)}$

From the classification of closed surfaces, we know that there exist closed curves $A_1, \ldots, A_g, B_1, \ldots, B_g$ such that their corresponding dual forms $\alpha_1, \ldots, \alpha_g, \beta_1, \ldots, \beta_g$ satisfy that

$$\int_M \alpha_i \wedge \alpha_j = \int_M \beta_i \wedge \beta_j = 0, \ \int_M \alpha_i \wedge \beta_j = \delta_{ij}.$$

Now for any smooth non-vanishing f on M, let

$$m_j = \frac{1}{2\pi i} \int_{A_j} \frac{\mathrm{d}f}{f}, \ n_j = \frac{1}{2\pi i} \int_{B_j} \frac{\mathrm{d}f}{f}$$

and consider the loop $\sigma = \prod_{j=1}^g A_j^{-n_j} B_j^{m_j}$, then for any closed curve τ we have

$$\int_M \frac{\mathrm{d}f}{f} \wedge \omega_\tau = 2\pi i \int_\sigma \omega_\tau.$$

Since $\alpha_1, \ldots, \alpha_g, \beta_1, \ldots, \beta_g$ form a basis of $H^1_{\mathrm{dR}}(M)$, we then conclude that for any closed form η on M we will have

$$\int_M \frac{\mathrm{d}f}{f} \wedge \eta = 2\pi i \int_\sigma \eta.$$

In particular, this means that $\dfrac{\partial f}{f}$ and $2\pi i \int_\sigma (-)$ are the same in $\left(\overline{H^0(K)}\right)^*$.

To sum up, we have proved via the Hodge decomposition theorem that

$$\left\{ \int_M \alpha \wedge (-) : \alpha = \frac{\partial f}{f} \in H^0(K) \right\} = \left\{ 2\pi i \int_\gamma (-) : \gamma \in \pi_1(M) \right\} \subset \left(\overline{H^0(K)}\right)^*,$$

which therefore implies that the moduli space of holomorphic line bundles of a given degree on M is given by

$$H^0(K)^* \Big/ H_1^{\mathrm{sing}}(M;\mathbb{Z}) =: J(M),$$

called the **Jacobian variety** of M. Our next goal is to show that $J(M)$ is actually a torus, i.e. $H_1(M;\mathbb{Z})$ acts as a lattice subgroup of $H^0(K)^* \cong \mathbb{C}^g$. To prove this we need the following theorem:

Theorem 3.5.3 (Riemann's bilinear relation)**.** *Let $A_1, \ldots, A_g, B_1, \ldots, B_g$ be a canonical choice of generators of $\pi_1(M)$, we have:*

- *Let α and β be smooth closed 1-forms on M, then*

$$\int_M \alpha \wedge \beta = \sum_{i=1}^g \int_{A_i} \alpha \cdot \int_{B_i} \beta - \int_{B_i} \alpha \cdot \int_{A_i} \beta.$$

3.6 The Kodaira embeddings

3.7 Automorphisms of a surface

Chapter 4

Other topics

4.1 Θ functions and the Lefschetz embedding

4.2 Countable topology

4.3 Gauge theory

Chapter 5

?

Theorem 5.0.1 (Example name)**.** *Example theorem content.*

Example reference to theorem 5.0.1 ————————

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